

Supermassive Black Hole Mergers



by Vikram Manikantan

Astro 300A: Dynamics, Spring 2026
University of Arizona

What's happening this week

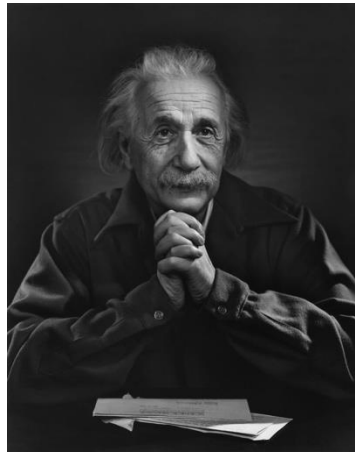
- Today (and some of Thursday):
 - I'm going to tell you about physical processes that drive supermassive black hole mergers
 - We're going to derive some key **timescales** that govern their evolution
- Thursday:
 - We'll talk about gravitational waves, and then you'll get to play with simulations/gravitational waveforms that we use to model **real black hole mergers**



Supermassive Black Hole Mergers



Qualitatively self-explanatory.
Quantitatively : $M_{\text{BH}} > 10^5 M_{\odot}$



Albert Einstein, credits:
Yousuf Karsh



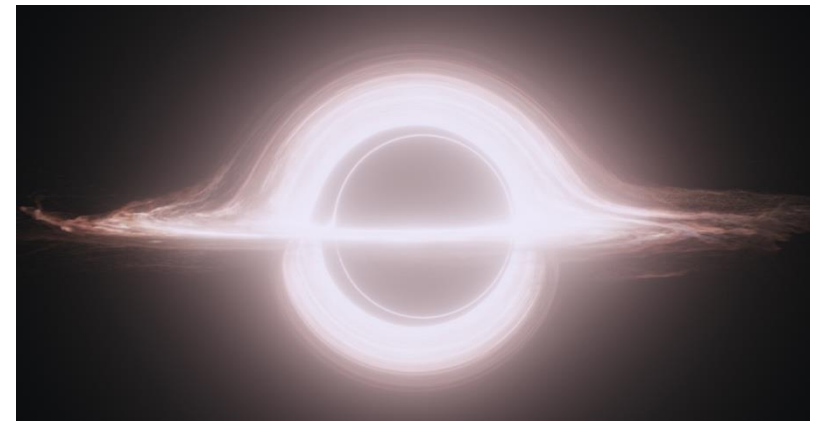
“a combination of two
things, especially
~~companies~~, into one”
- Oxford Dictionary

Black Holes – what are they?

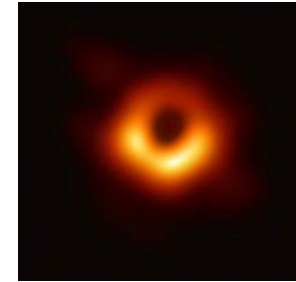
Come up to the board, write or draw, words or equations

- A region of spacetime separated by an event horizon
- What is an event horizon?
- Point at which no particle (not even photons) can escape to infinity¹

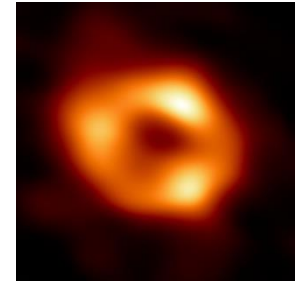
¹Spacetime and Geometry, Carroll



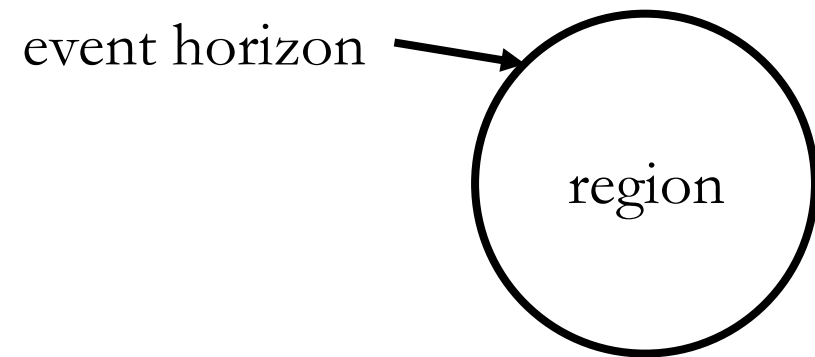
Interstellar, 2014

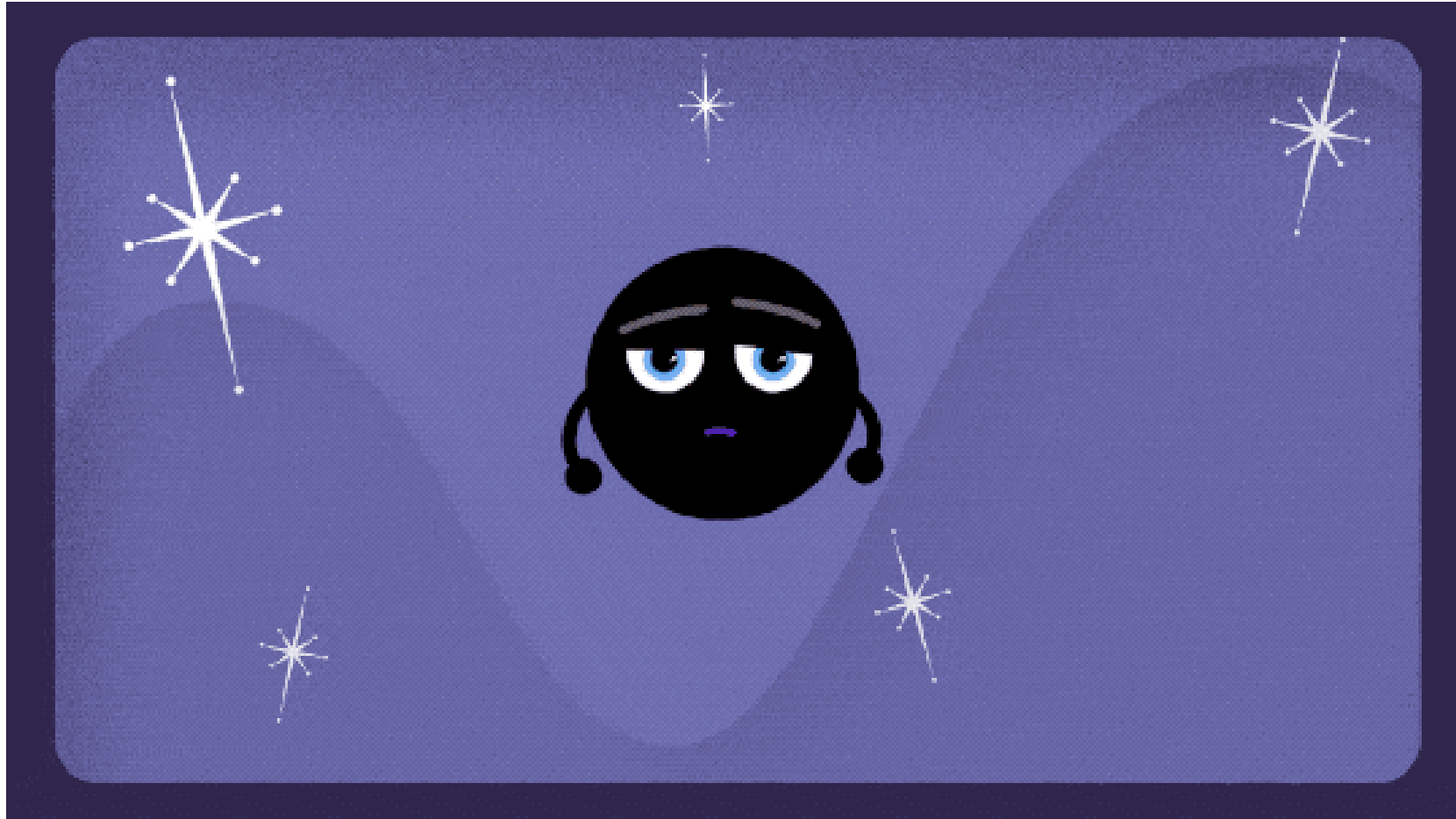


M87*, EHT



Sag A*, EHT





Credit: NASA's Goddard Space Flight Center

Supermassive Black Hole Mergers: what do we know?



1. Galaxies exist
2. They host central SMBHs
(usually)
3. They merge often



What we don't know



1. Galaxies exist
2. They host central SMBHs (usually)
3. They merge often

1. What happens to the central black holes? (BHs)
2. How do they merge?
3. And, what does it look like (astronomically)?

Our focus
this week

Concepts we are going to use:

1. Order-of-mag estimates (lectures 1-2)
2. Orbital dynamics
3. Dynamical Friction (lecture 9)
4. Gravitational waves (new!)

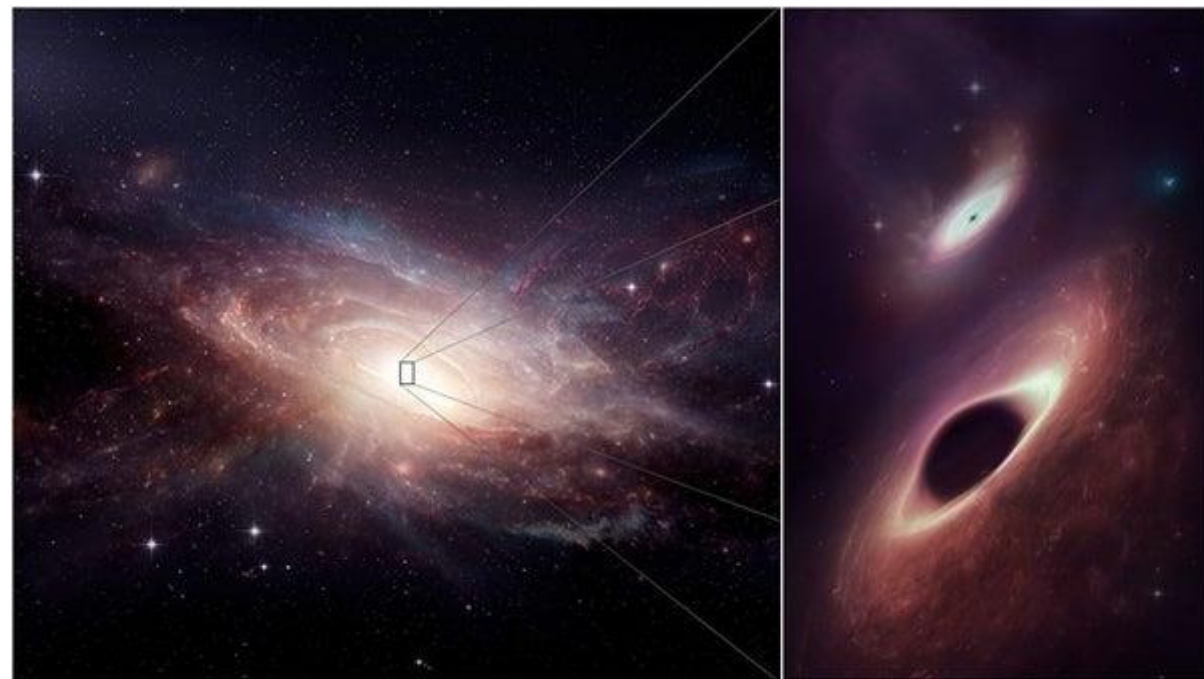
After this lecture, you will be able to:

1. **Describe** how dynamical friction and gravitational waves (GWs) both contribute to SMBH mergers
2. Use d/dt equations to **estimate timescales** for dynamic friction and GWs
3. **Compare** different timescales to **evaluate** when and which processes drive SMBHs to merger

* There is an in-class assignment today

Why do we care about SMBHs merging?

- Black holes are cool
- Where do SMBHs come from?
- SMBH mergers trace galaxy mergers throughout cosmic time
- Einstein's theory of GR is great, but *is it correct?*
- What about the behavior of particles in relativistic jets?
(Bogdanovic+2022, LISA Consortium+2023)



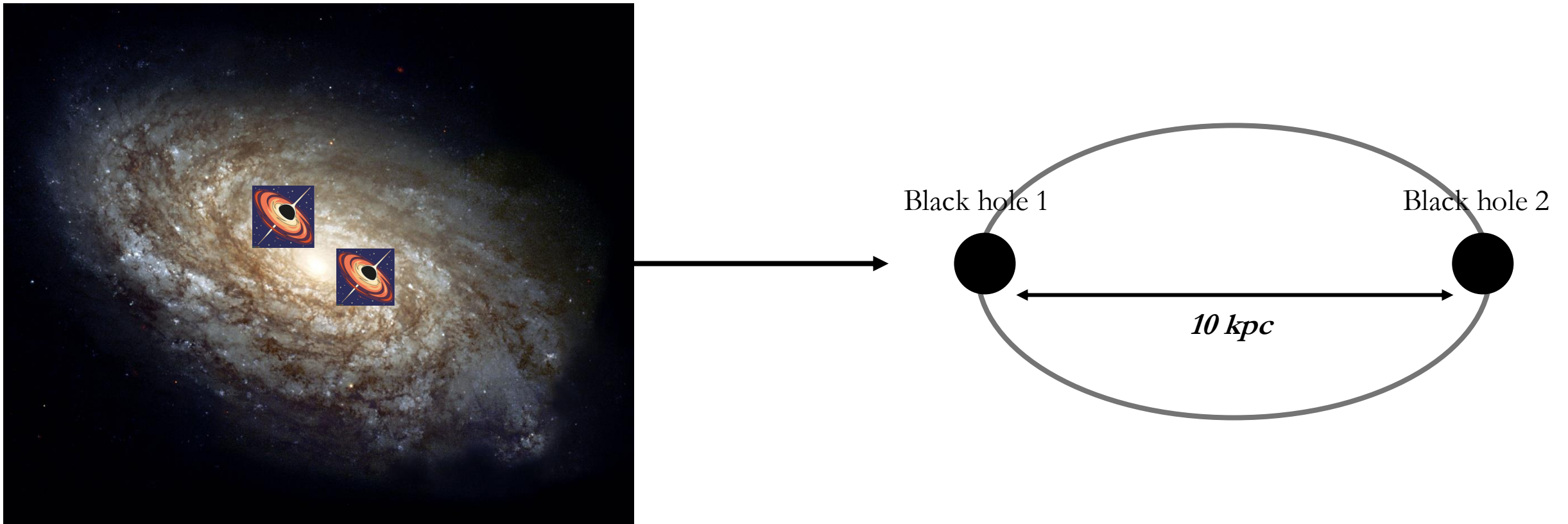
Simplify a complex problem

Learning goal: Compare different timescales to evaluate when and **which processes** drive SMBHs to merger



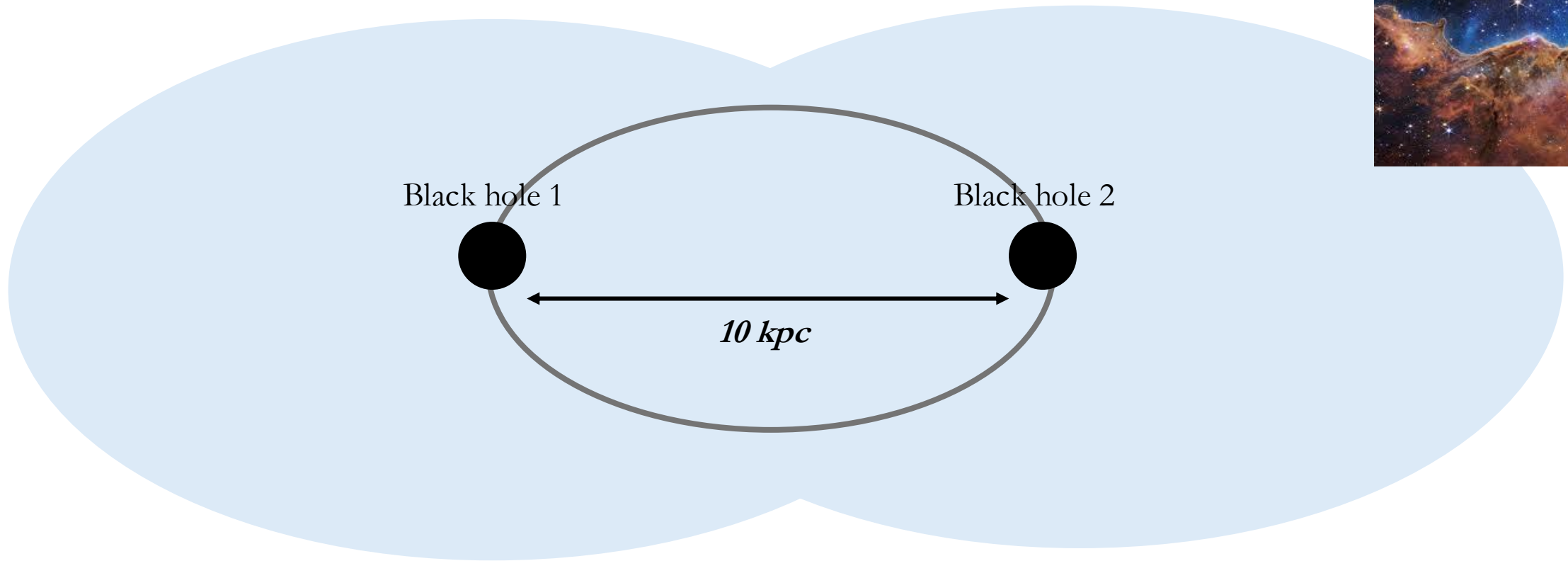
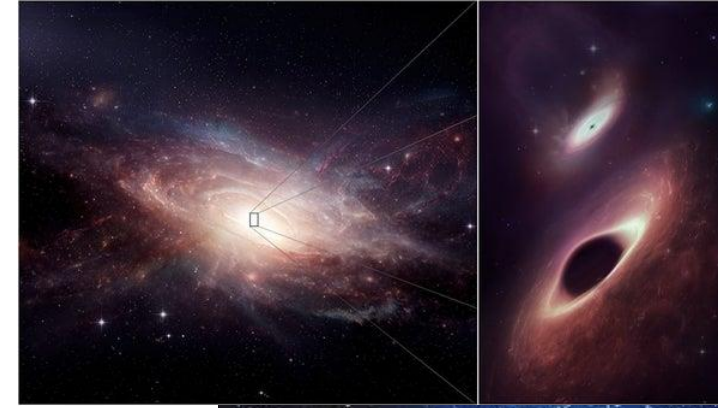
There's a lot going on here!
We need to identify what processes are happening

First, orbiting supermassive black holes



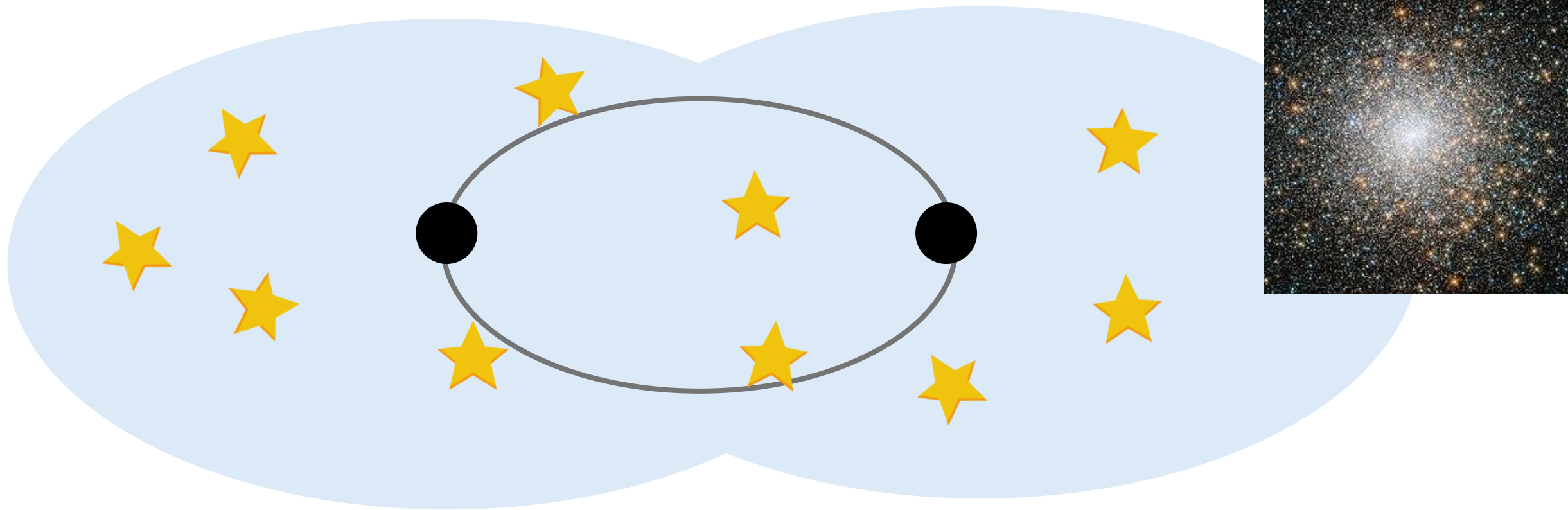
What other matter is there?

Interstellar Gas



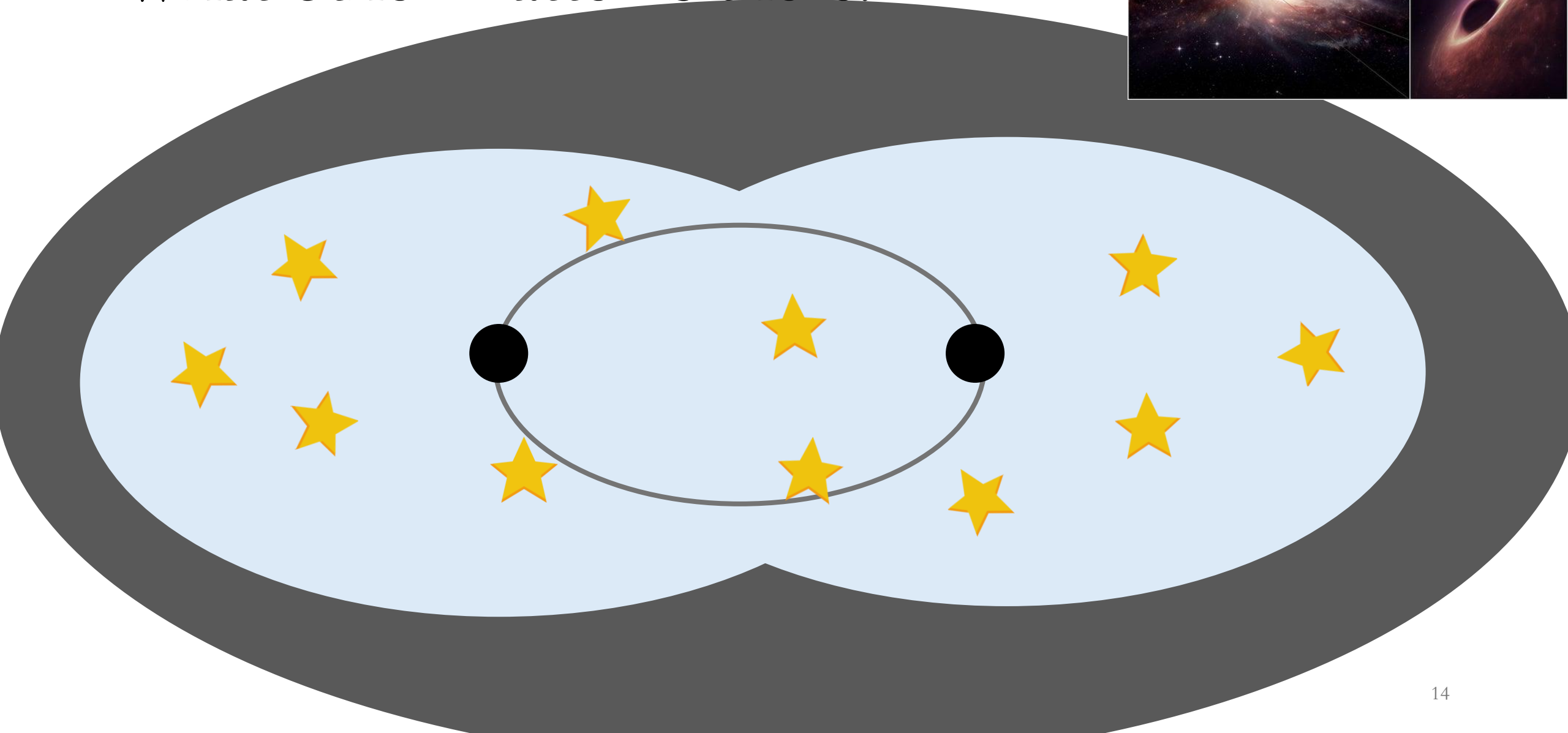
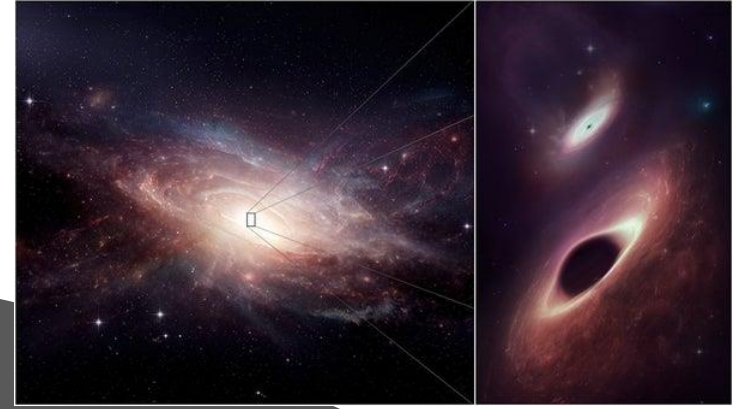
What other matter is there?

Stars



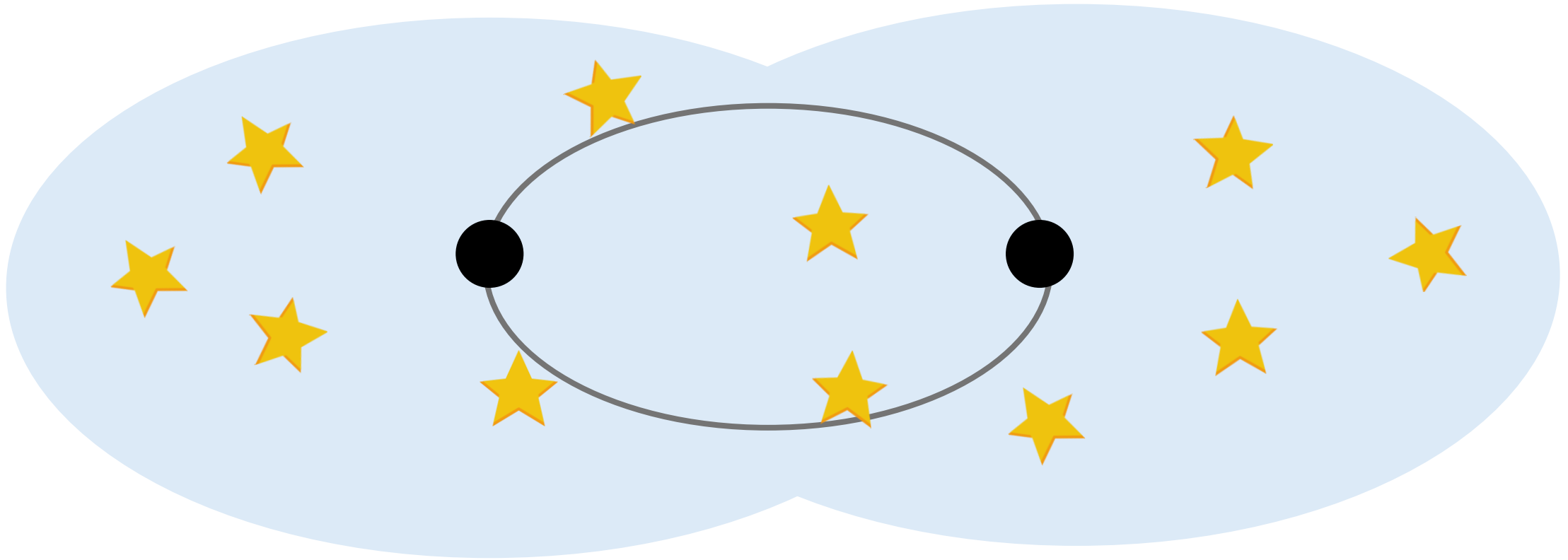
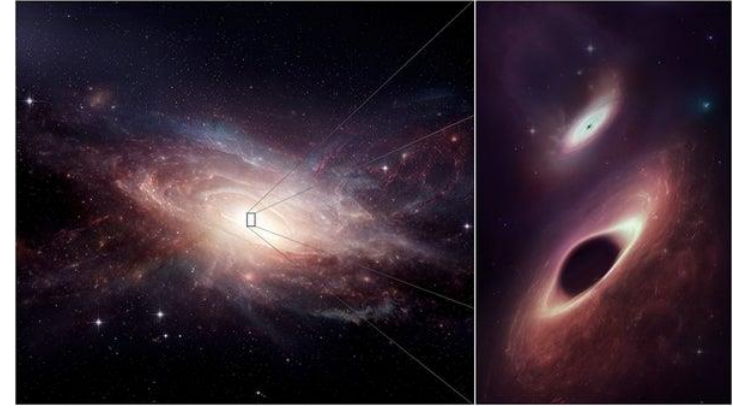
Dark Matter

What other matter is there?



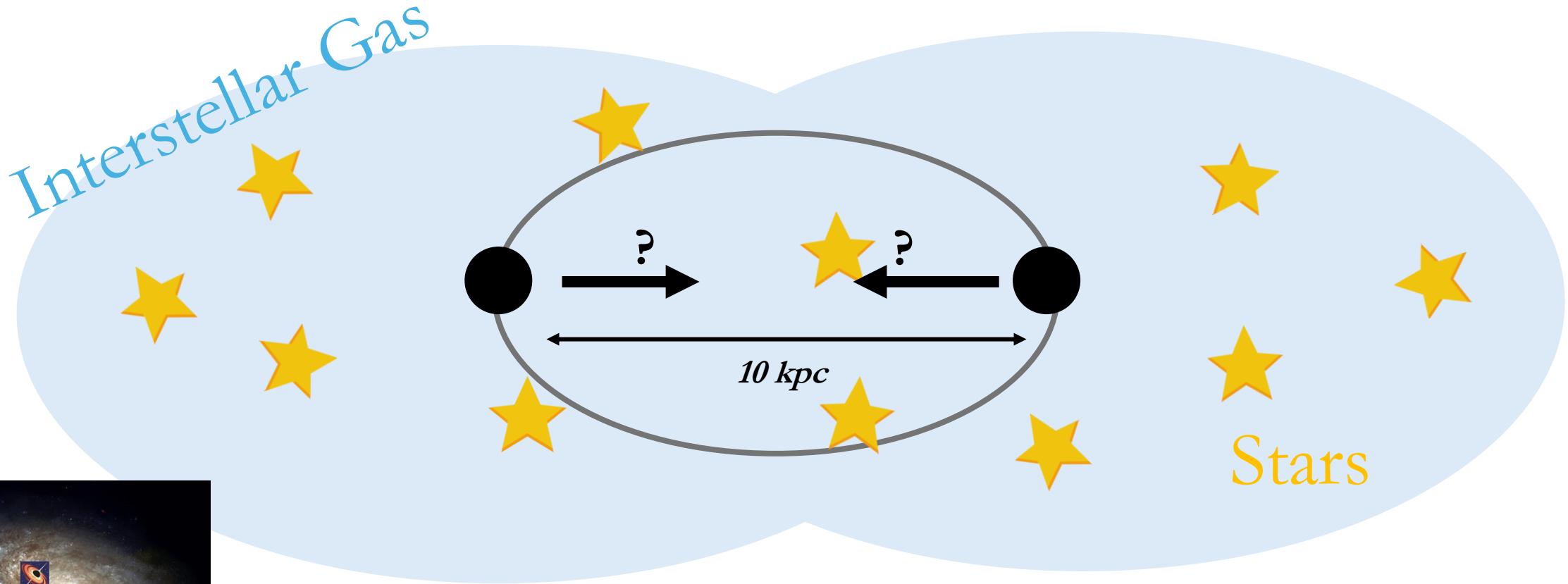
What other matter is there?

Dark Matter (assume not important)



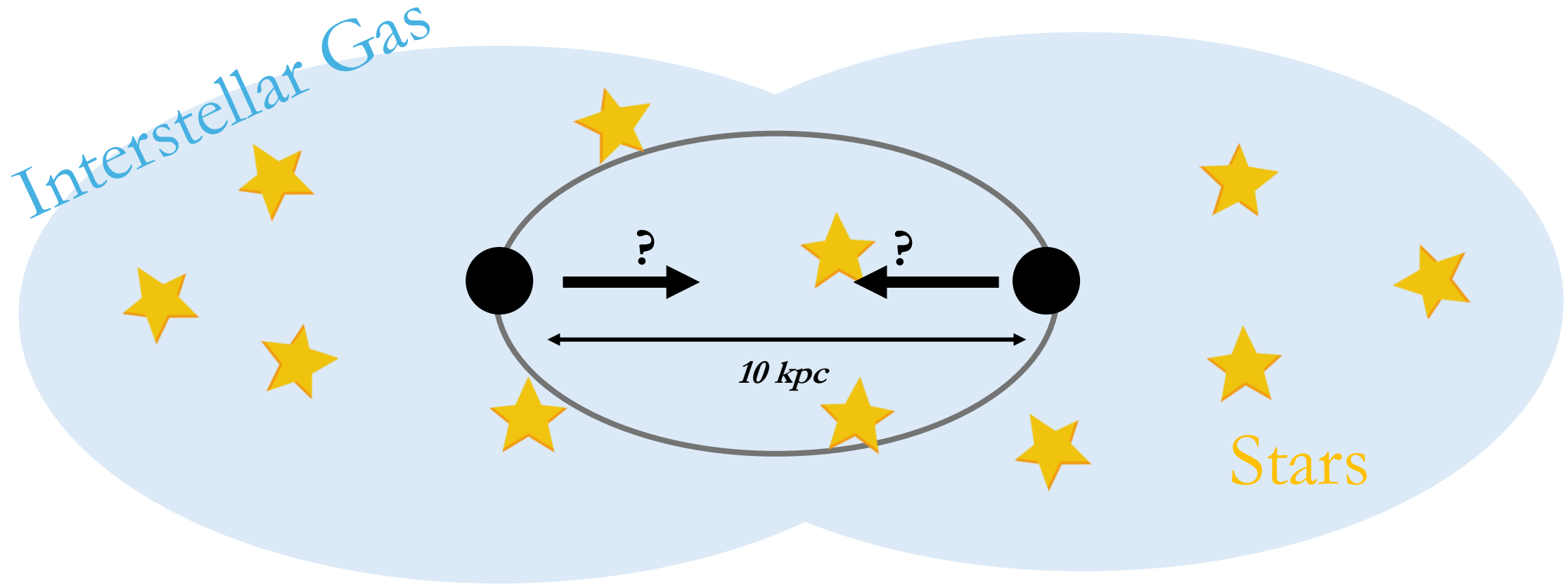
What needs to happen for these orbiting BHs to merge?

Discuss with your groups for 1 minute



What we know: there is gas, stars, and black holes

We want to get rid of angular momentum! How?

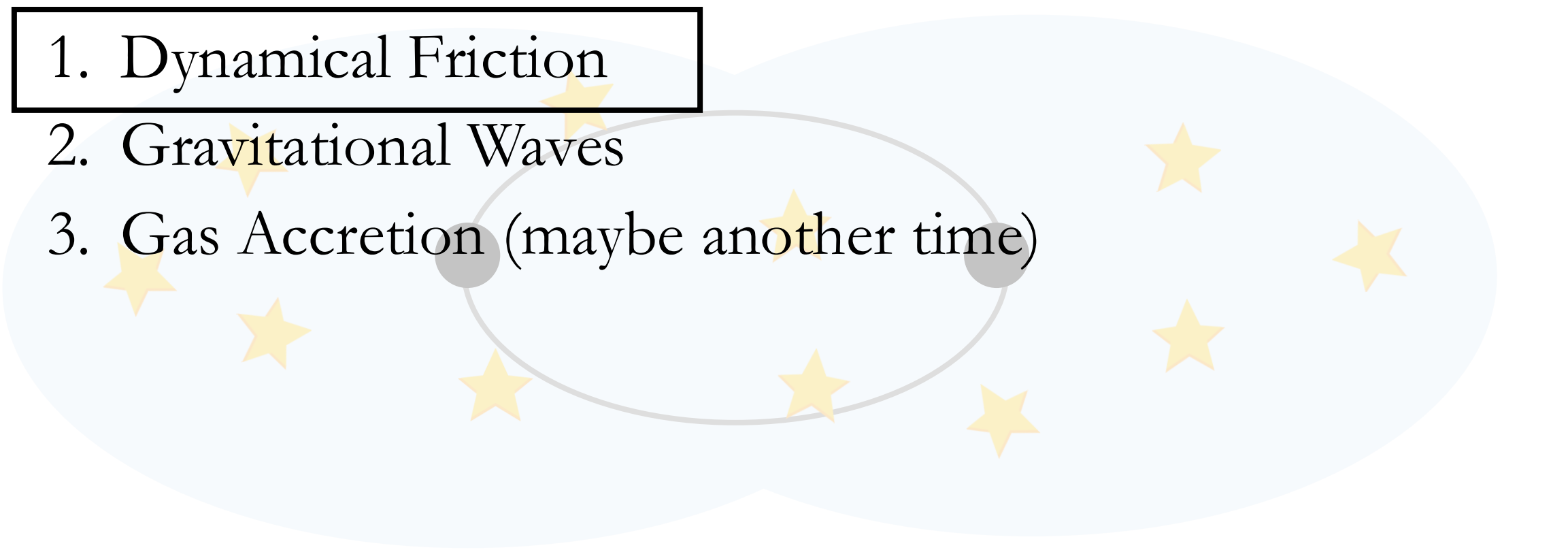


We want to get rid of angular momentum! How?

1. Dynamical Friction
2. Gravitational Waves
3. Gas Accretion (maybe another time)

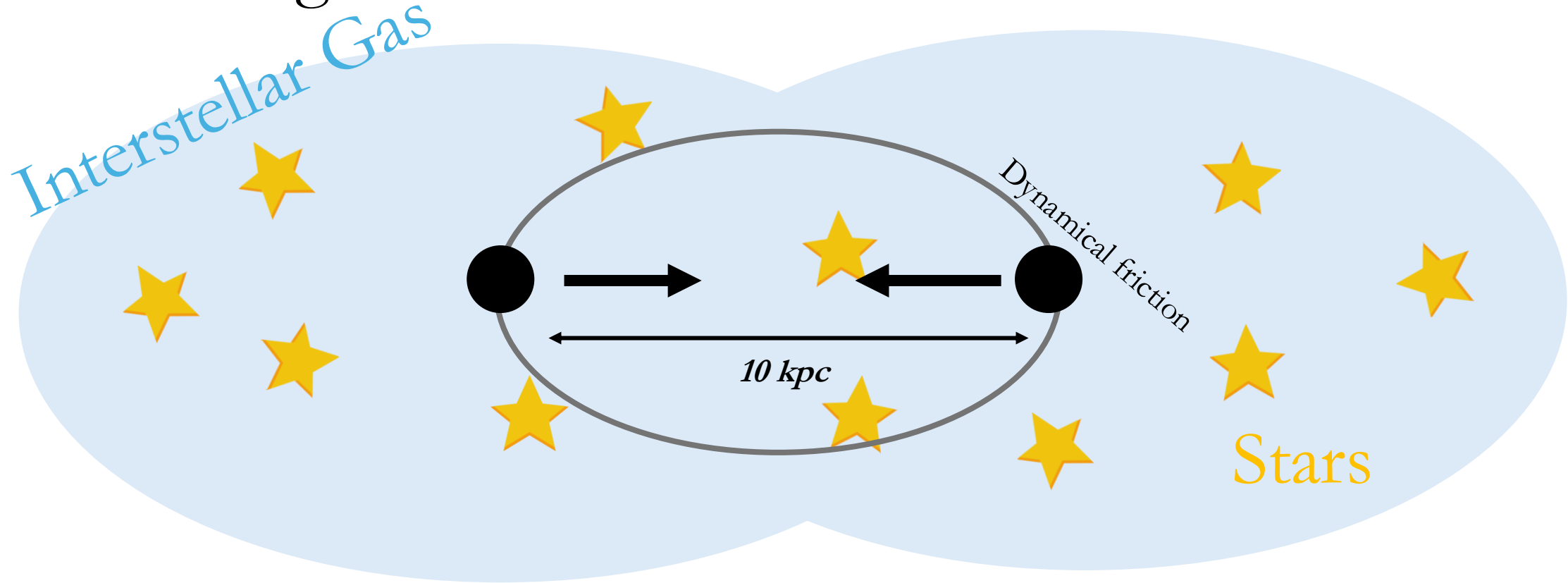
Not to scale

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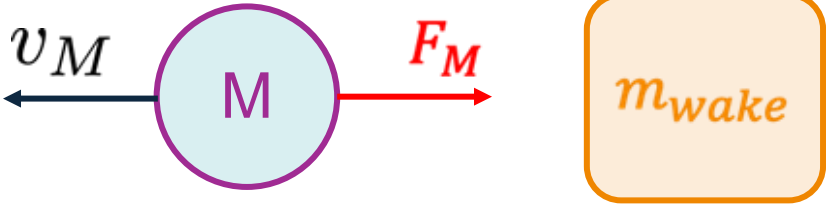
Not to scale

Our hypothesis: Dynamical friction through interactions with stars and gas will bring the black holes together



Recall Lecture 9:

Dynamical Friction

$$\frac{dv_M}{dt} \propto -\frac{G^2 M \rho}{v_M^2}$$


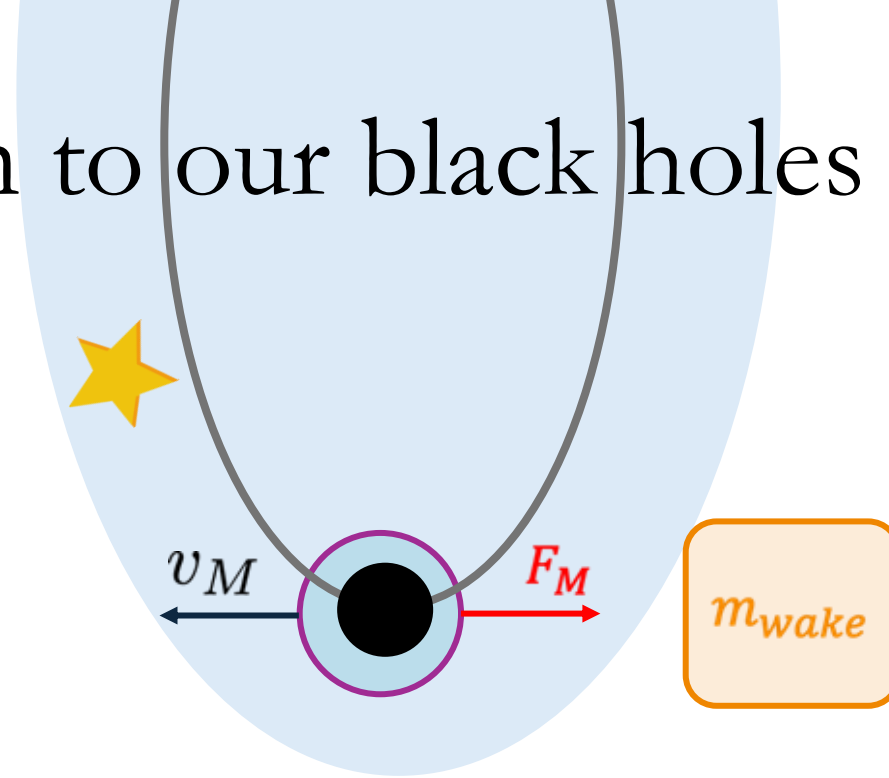
A diagram illustrating the concept of dynamical friction. A light blue circle labeled 'M' represents a massive object moving through a medium. A blue arrow labeled v_M points to the left, indicating the direction of motion. A red arrow labeled F_M points to the right, representing the friction force. To the right of the object is an orange rounded rectangle labeled m_{wake} , representing the wake mass.

$$\frac{d\vec{v}_M}{dt} = -\frac{4\pi \ln(\Lambda) G^2 \rho M}{v_M^2} \hat{v}_M$$

Applying this equation to our black holes

Dynamical Friction

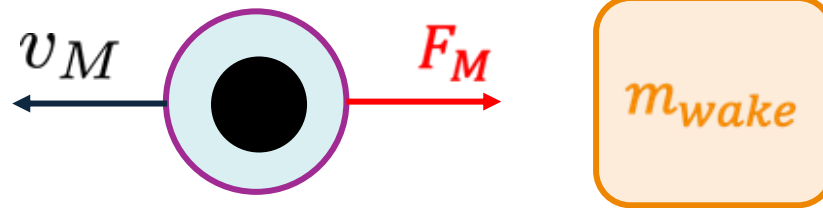
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$$\frac{d\vec{v}_M}{dt} = -\frac{4\pi \ln(\Lambda) G^2 \rho M}{v_M^2} \hat{v}_M$$

Applying this equation to our black holes

Dynamical Friction



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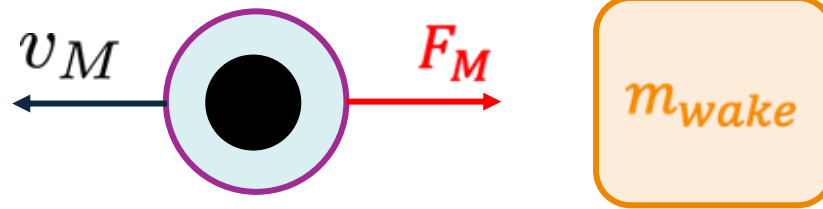
$$\frac{d\vec{v}_M}{dt} = -\frac{4\pi \ln(\Lambda) G^2 \rho M}{v_M^2} \hat{v}_M$$

Learning Goal: Use d/dt equations to **estimate timescales** for dynamic friction and GWs

Where is the time(scale) in this equation?

Now what do we want to do with the **time**?

Dynamical Friction



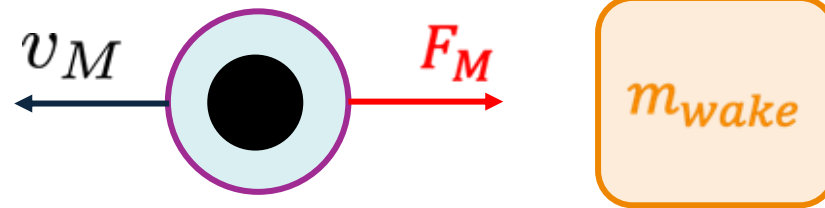
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Dynamical Friction



$$\frac{dv_M}{dt} \propto -\frac{G^2 M \rho}{v_M^2}$$

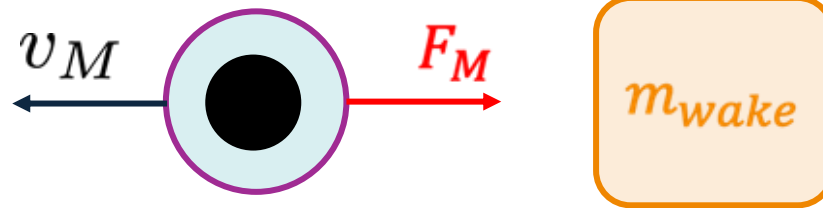
Learning Goal: Use d/dt equations to **estimate timescales** for dynamic friction and GWs

$$\frac{d\vec{v}_M}{dt} = -\frac{4\pi \ln(\Lambda) G^2 \rho M}{v_M^2} \hat{v}_M$$

We want to find the **time it takes** to slow the BHs down **to zero velocity**?

Now what do we want to do with the **time**?

Dynamical Friction



$$\frac{dv_M}{dt} \propto -\frac{G^2 M \rho}{v_M^2}$$

Learning Goal: Use d/dt equations to **estimate timescales** for dynamic friction and GWs

$$\frac{d\vec{v}_M}{dt} = -\frac{4\pi \ln(\Lambda) G^2 \rho M}{v_M^2} \hat{v}_M$$

We want to find the **time it takes** to slow the BHs down **to zero velocity**? **How?**

Hint, how do you get the distance travelled from this equation: $v = \frac{dx}{dt}$

Collect like terms and integrate both sides
(we're going to use this again)

$$\frac{d\vec{v}_M}{dt} = -\frac{4\pi \ln(\Lambda) G^2 \rho M}{v_M^2} \hat{v}_M$$

$$\Rightarrow \int_{v_0}^0 v^2 dv = - \int_{?}^{?} 4\pi \ln(\Lambda) G^2 \rho M dt$$

What should these bounds be?

Discuss with your partners and
answer on your boards

A. T_{stop} to 0

B. 0 to M .

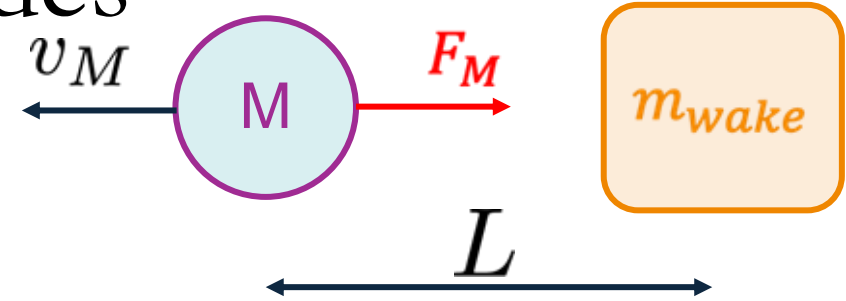
C. 0 to T_{stop}

D. T_i to T_f

Recall our goal:

How do we find the
time it takes to slow
the BHs down to zero
velocity?

Re-arrange and integrate both sides
(we're going to use this again)



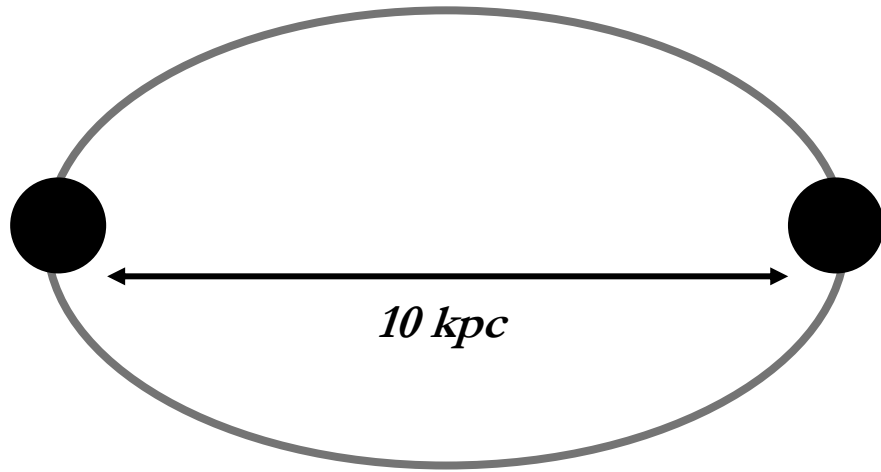
$$\frac{d\vec{v}_M}{dt} = -\frac{4\pi \ln(\Lambda) G^2 \rho M}{v_M^2} \hat{v}_M$$

$$\Rightarrow \int_{v_0}^0 v^2 dv = - \int_0^{T_{stop}} 4\pi \ln(\Lambda) G^2 \rho M dt$$

$$\Rightarrow -\frac{1}{3} v_0^3 = -4\pi \ln(\Lambda) G^2 \rho M T_{stop}$$

$$\Rightarrow T_{stop} = \frac{v_M^3}{12\pi \ln \Lambda G^2 \rho M}$$

How long does it take dynamical friction to bring our black holes to a stop?



- Assuming the total binary mass is $M = 10^7 M_{\text{sun}}$
- and they have orbital separation $a = 10 \text{ kpc}$,
- Using Kepler's laws, their orbital velocities are, $v \sim 2 \text{ km/s}$

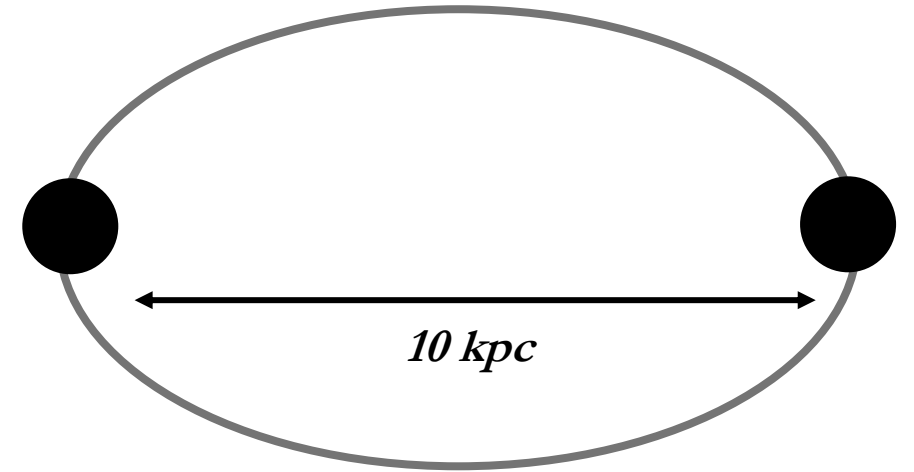
$$\Rightarrow T_{\text{stop}} = \frac{v_M^3}{12\pi \ln \Lambda G^2 \rho M}$$

Plugging in the parameters

$$\Rightarrow T_{stop} = \frac{v_M^3}{12\pi(10)G^2\rho M}$$

$$\Rightarrow T_{stop} \sim 0.00095 \text{ years} \sim 8.4 \text{ hours}$$

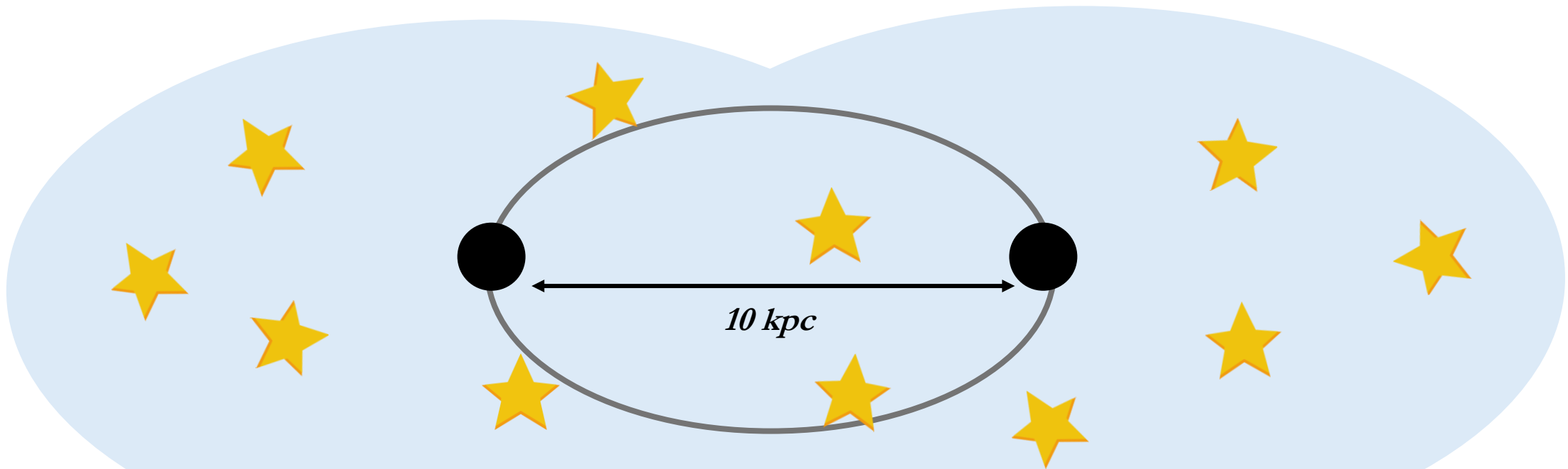
- $v \sim 2 \text{ km/s}$
- $\rho \sim 10^{-14} \text{ kg/m}^3$
- $M \sim 10^7 \text{ Msun}$
- $\ln \Lambda \sim 10$



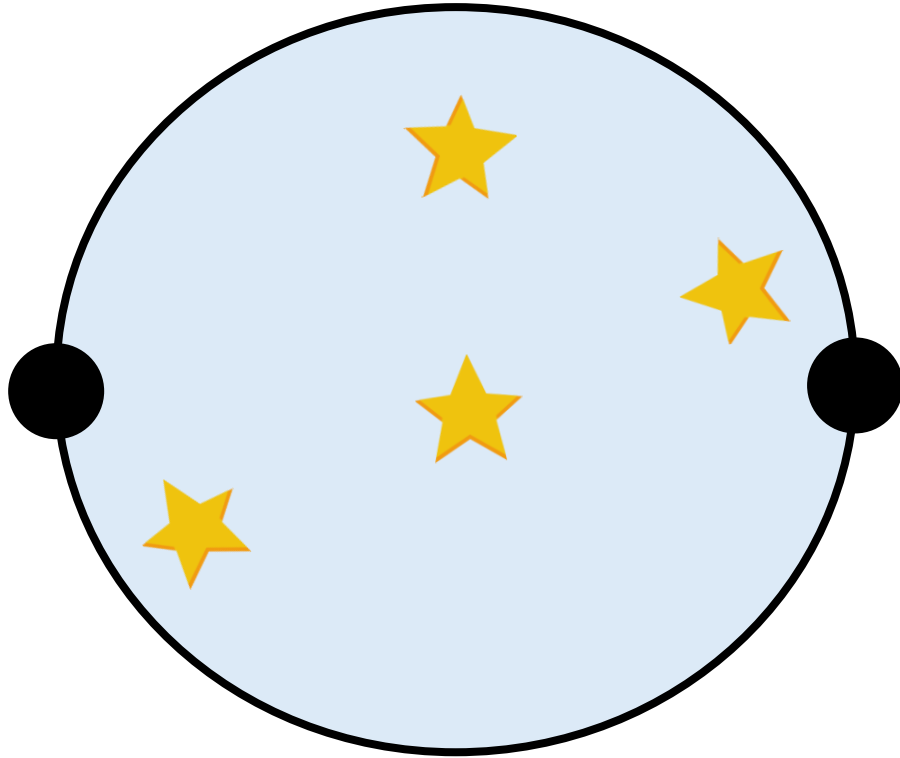
Done! Or are we? Discuss with your group:

1. Does this timescale look correct? (Yes/no)
2. Why or why not? (1 minute write on your board)

$$\Rightarrow T_{stop} = \frac{v_M^3}{12\pi(10)G^2\rho M} \sim 0.00095 \text{ years} \sim 8.4 \text{ hours}$$



1. This is an incomplete picture! We used the wrong mass



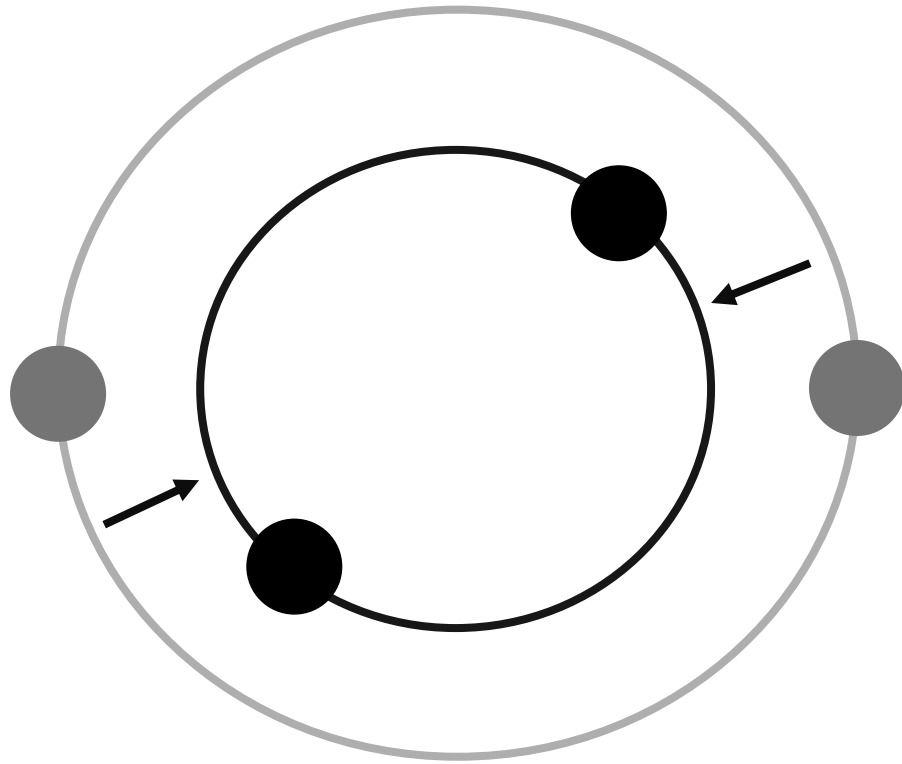
- The mass of this system is not just the BH mass!
- Remember enclosed mass from HW6?
- The gravitational mass is really the mass of the galaxies:

$$M_{MW} = 10^{12} M_{\text{sun}}$$

- Recalculating the Keplerian velocity:
 $v = 600 \text{ km/s}$
- $T_{\text{stop}} \sim (600/2)^3 * 8 \text{ hours} \sim 25,000 \text{ yrs}$

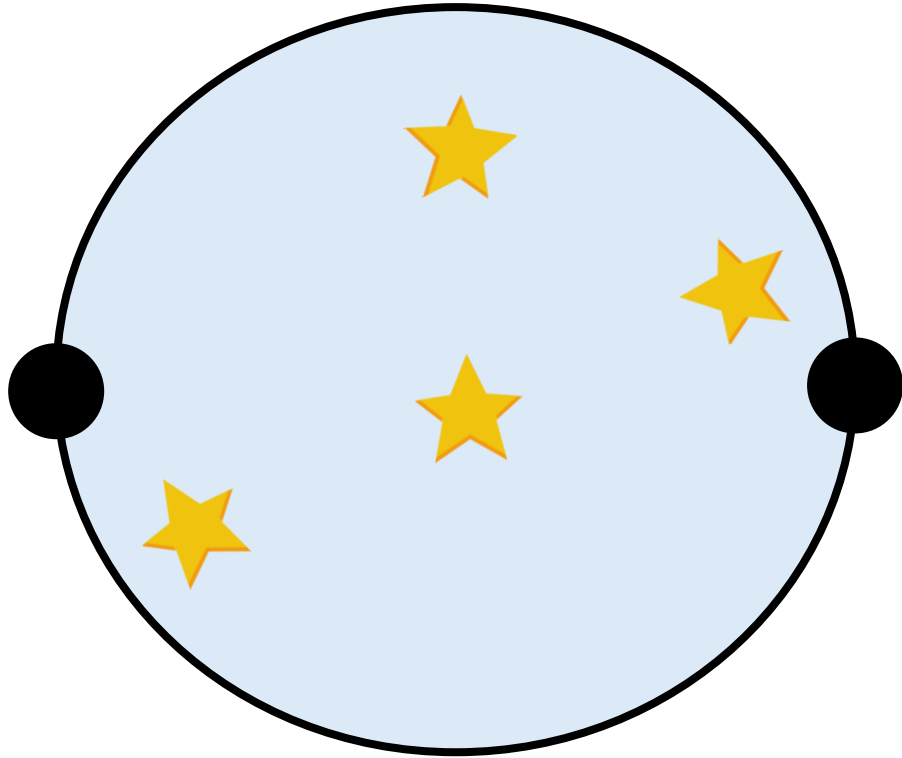
$$\Rightarrow T_{\text{stop}} = \frac{v_M^3}{12\pi(10)G^2\rho M}$$

2. This is an incomplete picture! “Slowing them down” does not bring them to merger



- Slowing down leads to a new orbital separation, not to merger!
- The BHs will speed up again, according to Kepler's law
- **So, we need to think of a more complete description of angular momentum loss**

We've already used dv/dt , what else can we do?

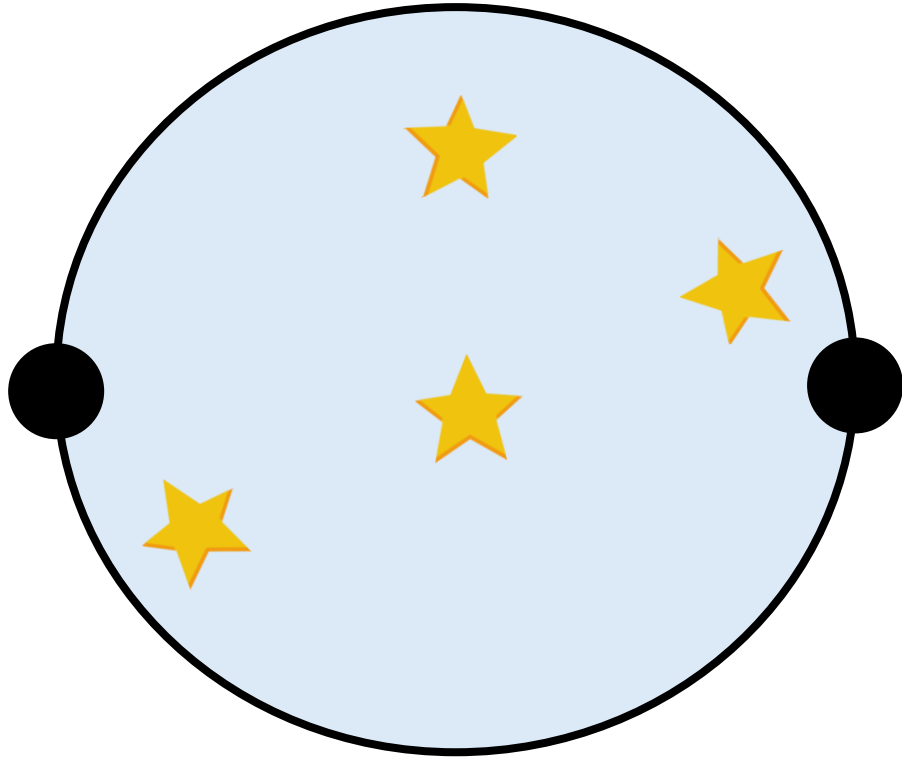


- Instead of velocity, let's think about angular momentum

$$\frac{d\vec{v}_M}{dt} = -\frac{4\pi \ln(\Lambda) G^2 \rho M}{v_M^2} \hat{v}_M$$

$$\Rightarrow T_{stop} = \frac{v_M^3}{12\pi \ln \Lambda G^2 \rho M}$$

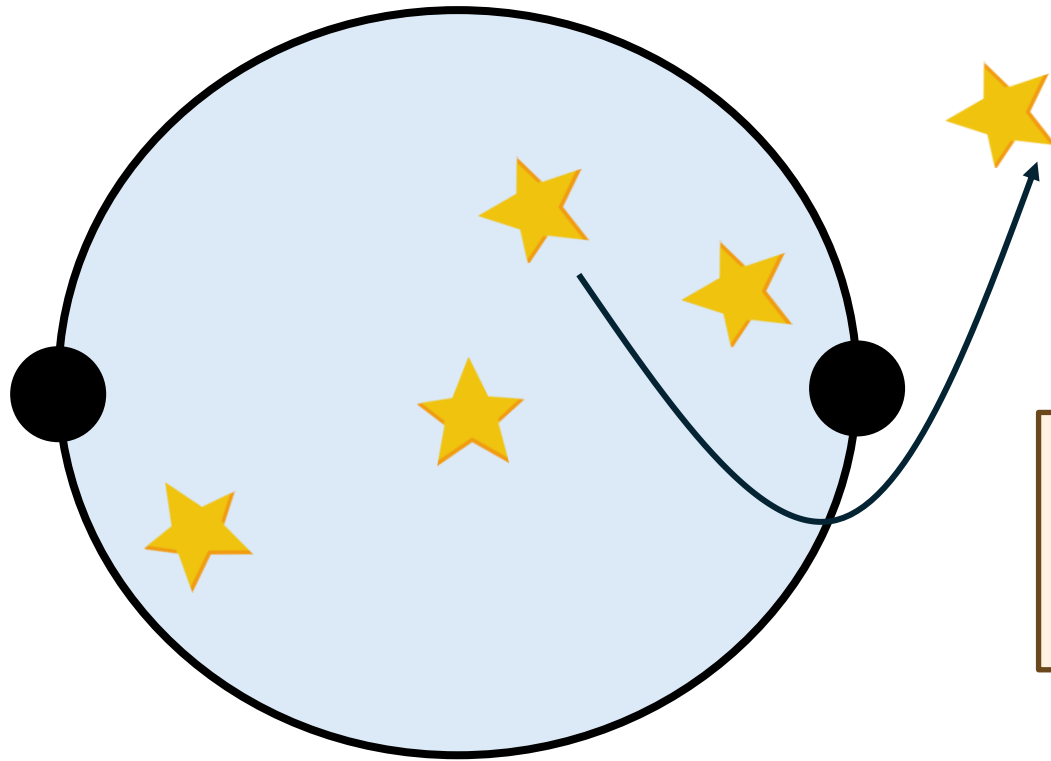
How do we quantify angular momentum loss?



- Define an approximate (strong emphasis on $\sim$) $\mathbf{dL}/dt$

$$\frac{dL}{dt} \sim ??$$

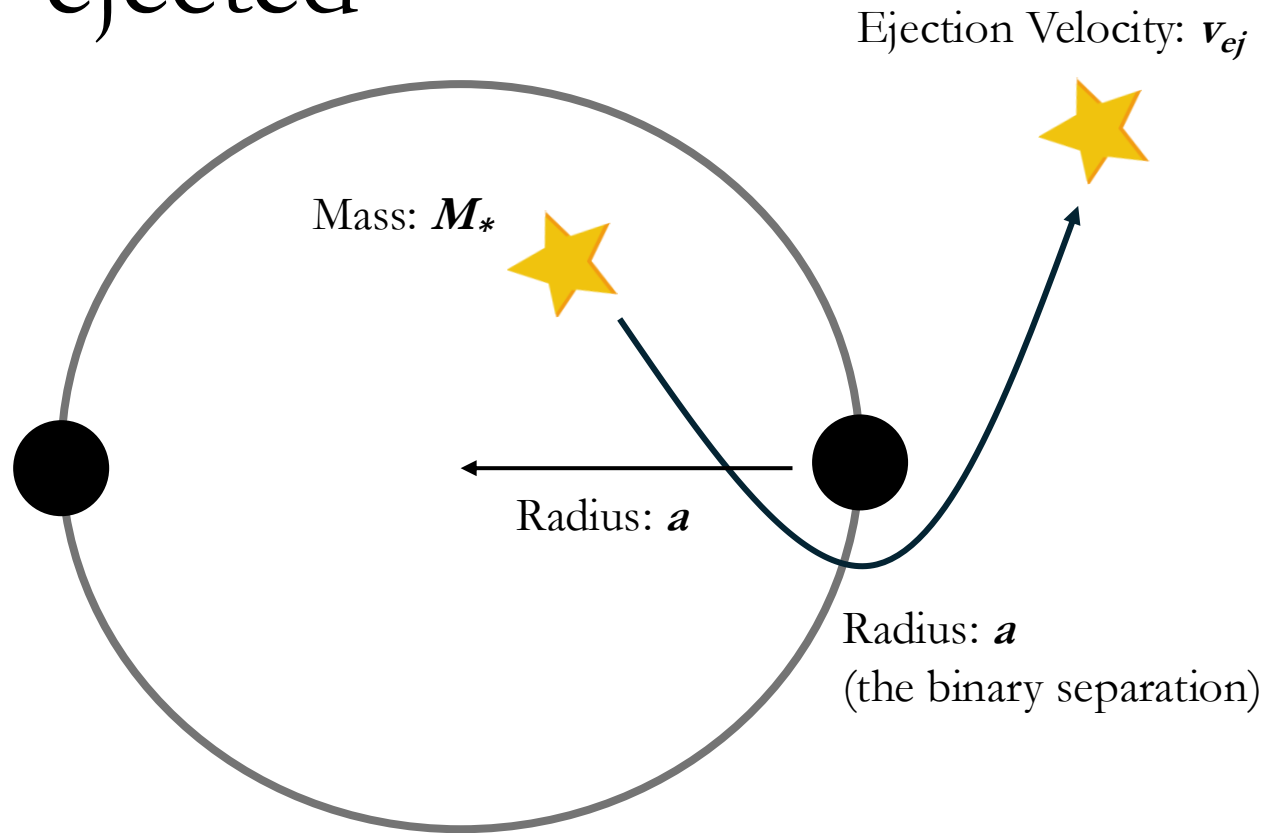
How do we quantify angular momentum loss?



- Define an approximate (strong emphasis on $\sim\sim$) $d\mathbf{L}/dt$

$$\frac{dL}{dt} = \Delta L / \Delta t$$

Assume a star interacts with the binary, and is ejected



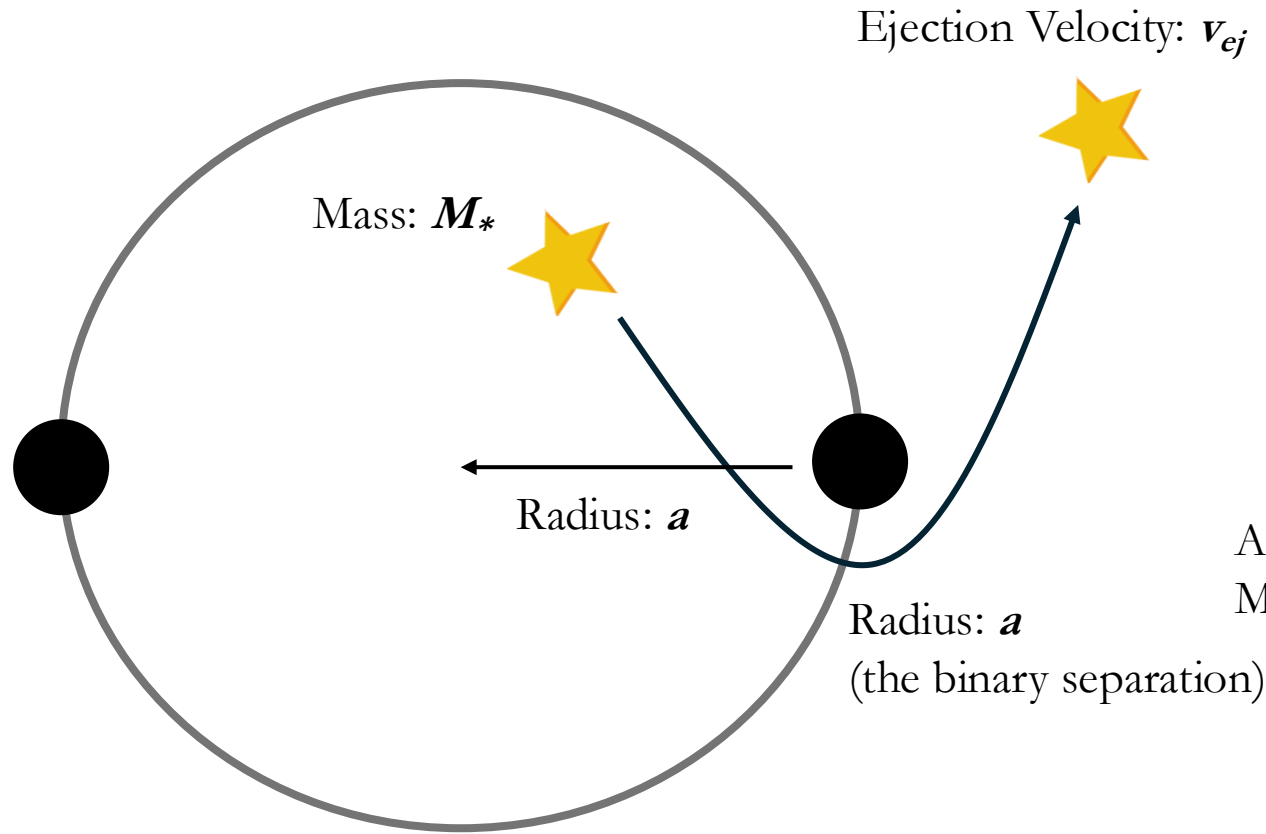
$$\frac{dL}{dt} = \frac{\Delta L}{\Delta t}$$

$$\Delta L = ?$$

$$\Delta L = M_* v_{ej} a$$

REMEMBER: this is for **one** stellar interaction

Generalize to a rate of stellar interactions



$$\frac{dL}{dt} \sim \frac{\Delta L}{\Delta t}$$

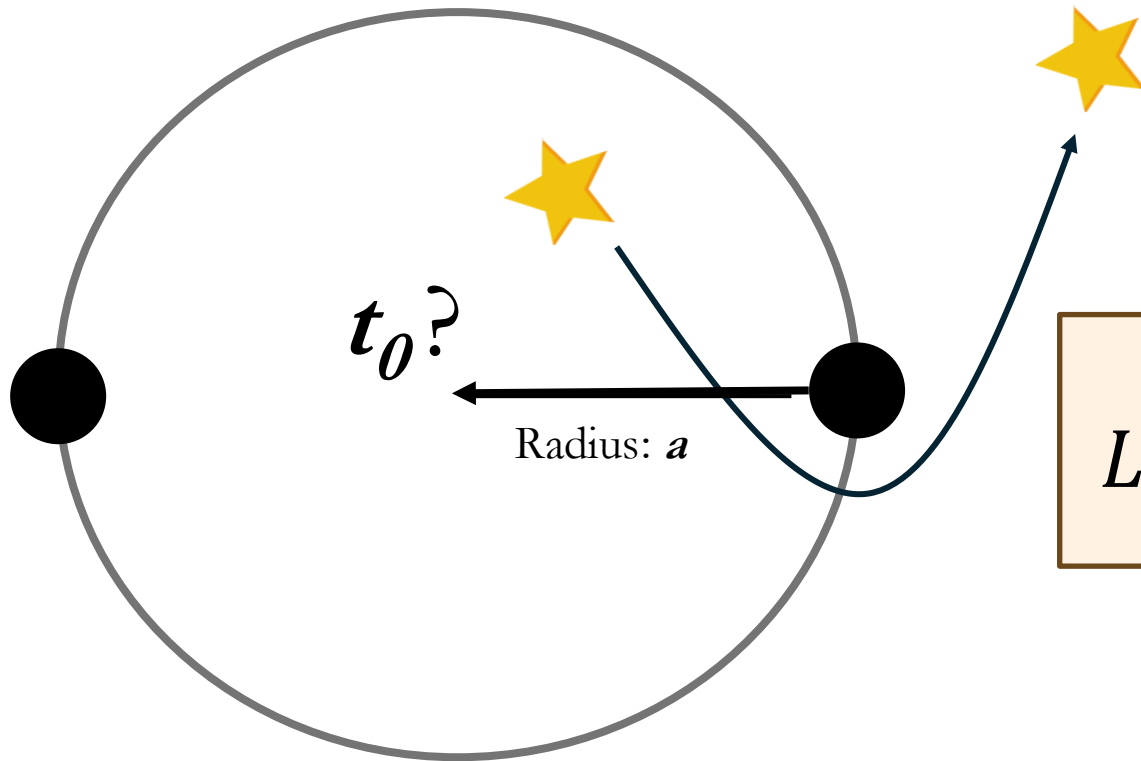
$$\Delta L_{one} = -M_* v_{ej} a$$

Assume n_{int} stellar interactions per year, and let's make the Mass and velocity averages, such that:

$$\frac{dL}{dt} = -\bar{M}_* \bar{v}_{ej} a n_{int}$$

The units check out!

Now, find time it takes to get $\mathbf{L}$ to $\mathbf{0}$



$$\frac{dL}{dt} = -\bar{M}_* \bar{v}_{ej} a n_{int}$$

Not a function of L ...

$$L = M\sqrt{GMa} \Rightarrow a = \frac{L^2}{GM^3}$$

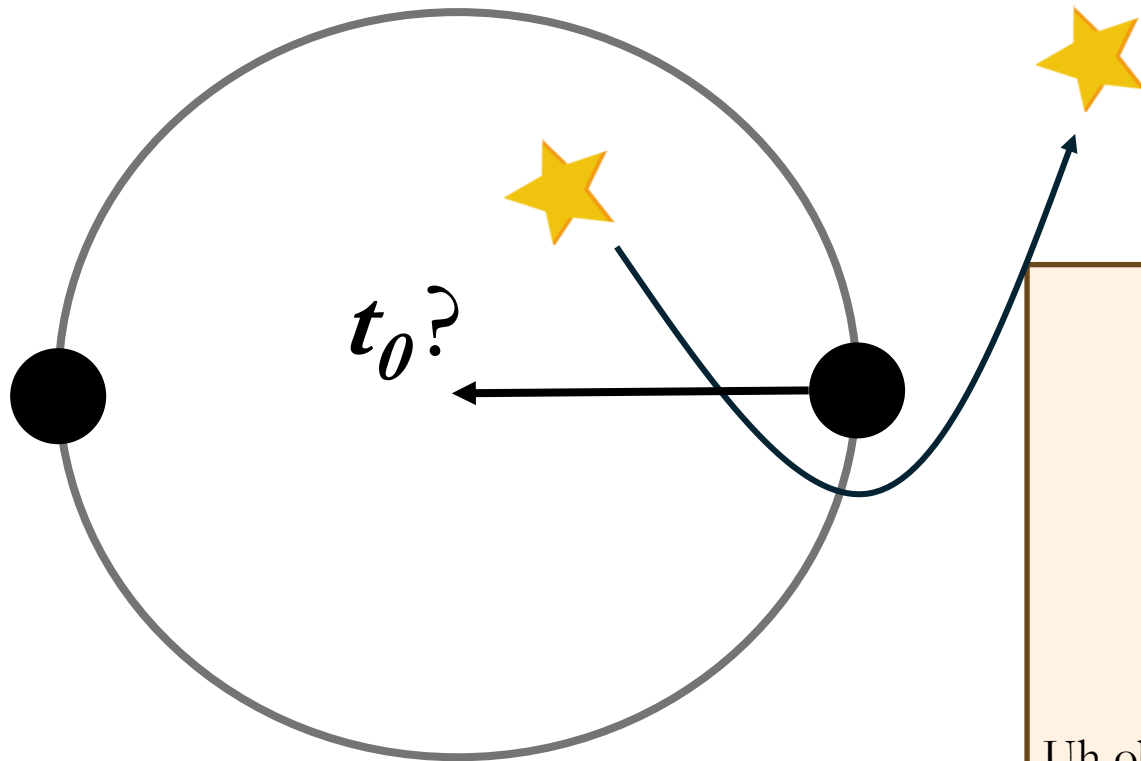
$$t_0 = ?$$

Recall from earlier:

$$\frac{d\vec{v}_M}{dt} = -\frac{4\pi \ln(\Lambda) G^2 \rho M}{v_M^2} \hat{v}_M$$

$$\Rightarrow T_{stop} = \frac{v_M^3}{12\pi \ln \Lambda G^2 \rho M}$$

Now, find time it takes to get $\mathbf{L}$ to some $\mathbf{0}$



$$\frac{dL}{dt} = -\bar{M}_* \bar{v}_{ej} \frac{L^2}{GM^3} n_{int}$$

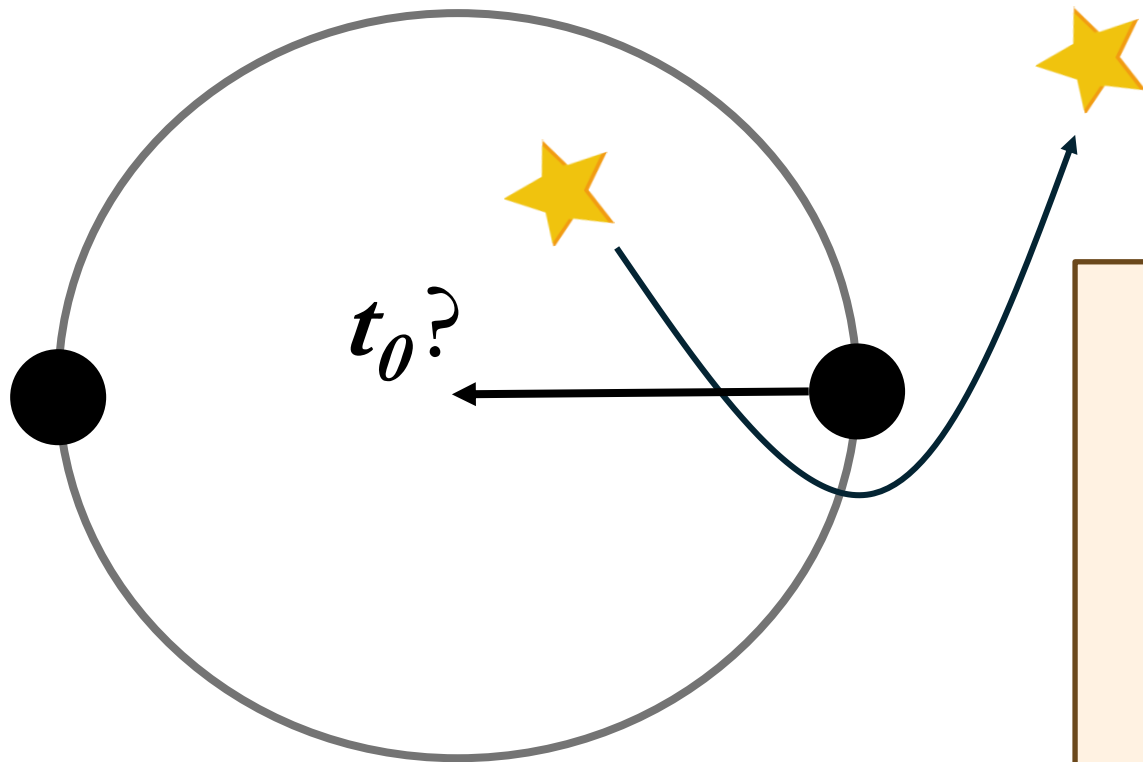
$$dL = -\bar{M}_* \bar{v}_{ej} \frac{L^2}{GM^3} n_{int} dt$$

Collect like terms and integrate both sides:

$$\int_{L_0}^0 \frac{1}{L^2} dL = \int_0^{t_0} \frac{\bar{M}_* \bar{v}_{ej} n_{int}}{GM^3} dt$$

Uh oh $\left[-\frac{1}{L} \right]_{L_0}^0 = -\frac{\bar{M}_* \bar{v}_{ej} n_{int}}{GM^3} t_0$

Now, find time it takes to get $\mathbf{L}$ to some $\mathbf{L}_{min}$



$$\frac{dL}{dt} = -\bar{M}_* \bar{v}_{ej} \frac{L^2}{GM^3} n_{int}$$

$$dL = -\bar{M}_* \bar{v}_{ej} \frac{L^2}{GM^3} n_{int} dt$$

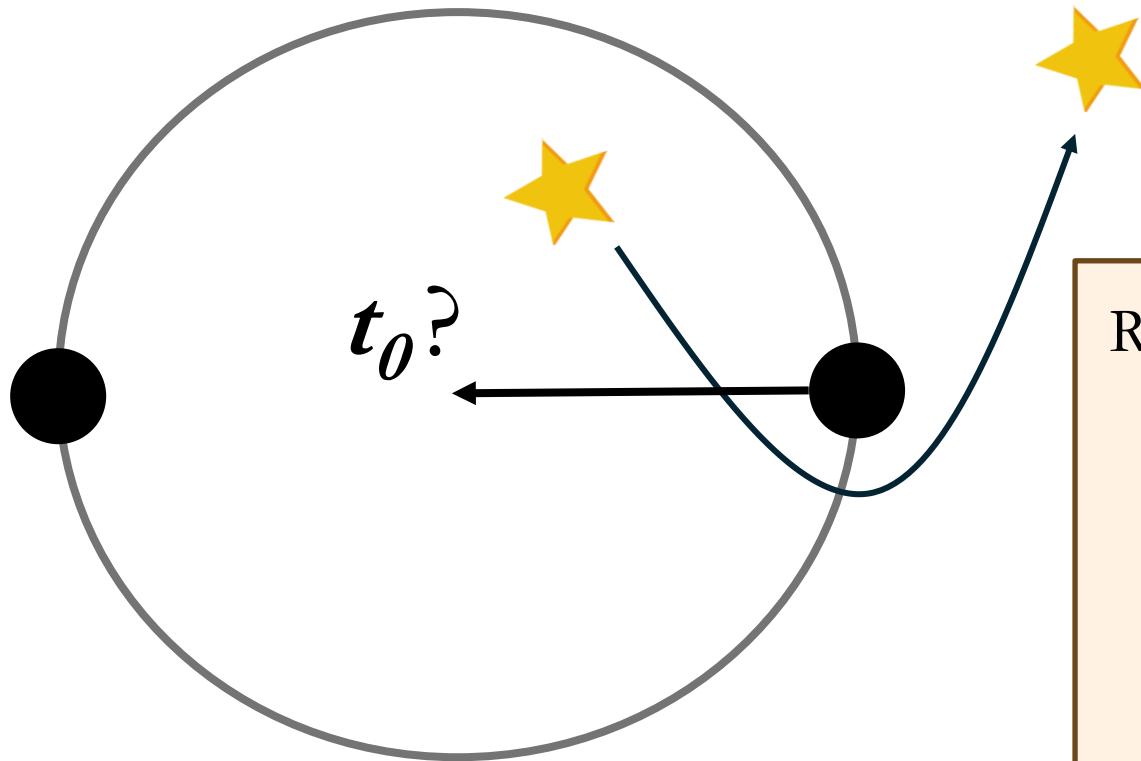
Collect like terms and integrate both sides:

$$\int_{L_0}^0 \frac{1}{L^2} dL = \int_0^{t_0} \frac{\bar{M}_* \bar{v}_{ej} n_{int}}{GM^3} dt$$

$$\left[-\frac{1}{L} \right]_{L_0}^{L_{min}} = -\frac{\bar{M}_* \bar{v}_{ej} n_{int}}{GM^3} t_0$$

$$\left(\frac{1}{L_0} - \frac{1}{L_{min}} \right) = -\frac{\bar{M}_* \bar{v}_{ej} n_{int}}{GM^3} t_0$$

Now, find time it takes to get L to some L_{min}



$$\frac{dL}{dt} = -\bar{M}_* \bar{v}_{ej} \frac{L^2}{GM^3} n_{int}$$

Re-arrange for t_0 :

$$\left(\frac{1}{L_0} - \frac{1}{L_{min}} \right) = -\frac{\bar{M}_* \bar{v}_{ej} n_{int}}{GM^3} t_0$$

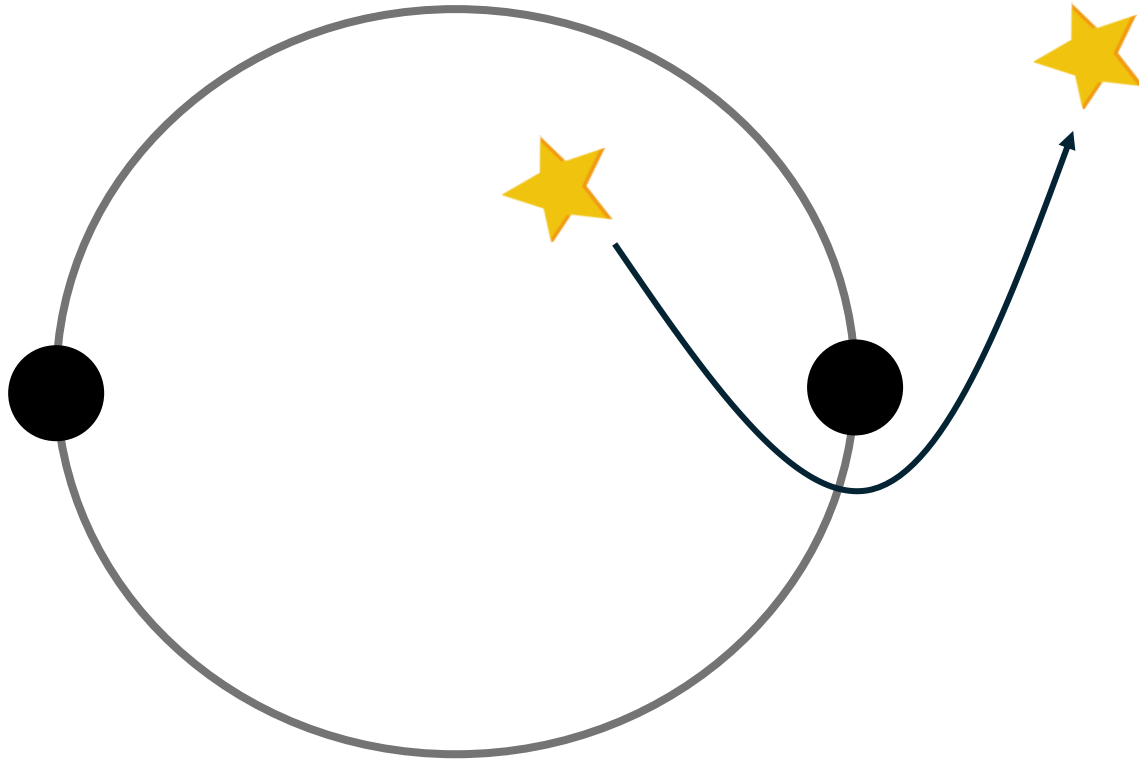
$$\Rightarrow t_0$$

$$= \frac{GM^3}{\bar{M}_* \bar{v}_{ej} n_{int}} \left(\frac{1}{L_{min}} - \frac{1}{L_0} \right)$$

We can make a few more simplifications, like replacing L with known quantities:

$$L = M \sqrt{GMa}$$

We did it?



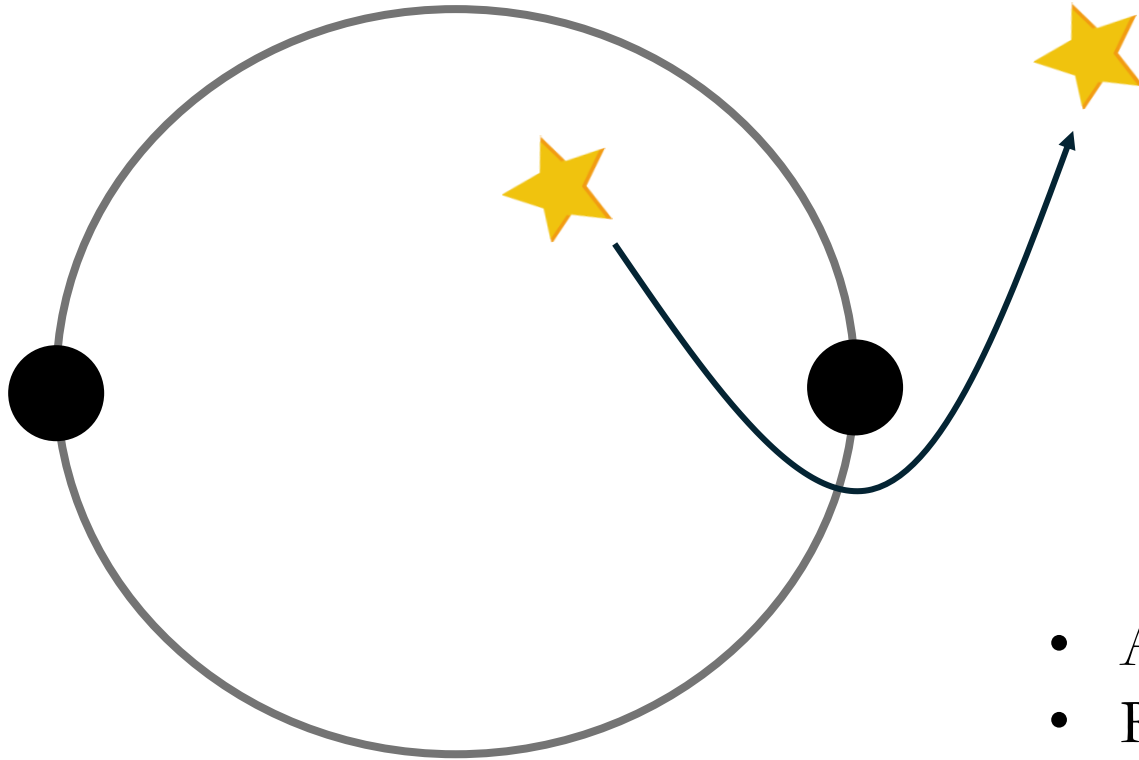
$$\frac{dL}{dt} = -\bar{M}_* \bar{v}_{ej} \frac{L^2}{GM^3} n_{int}$$

$$\Rightarrow t_0 = \frac{\sqrt{GM^3}}{\bar{M}_* \bar{v}_{ej} n_{int}} \left(\frac{1}{\sqrt{a_{min}}} - \frac{1}{\sqrt{a_0}} \right)$$

- Are the units correct?
- Do the proportionalities make sense?

a lot is hidden in here, but we can say that
no. interactions is proportional to stellar
density: $n_{int} \propto \rho_*$

We did it...?

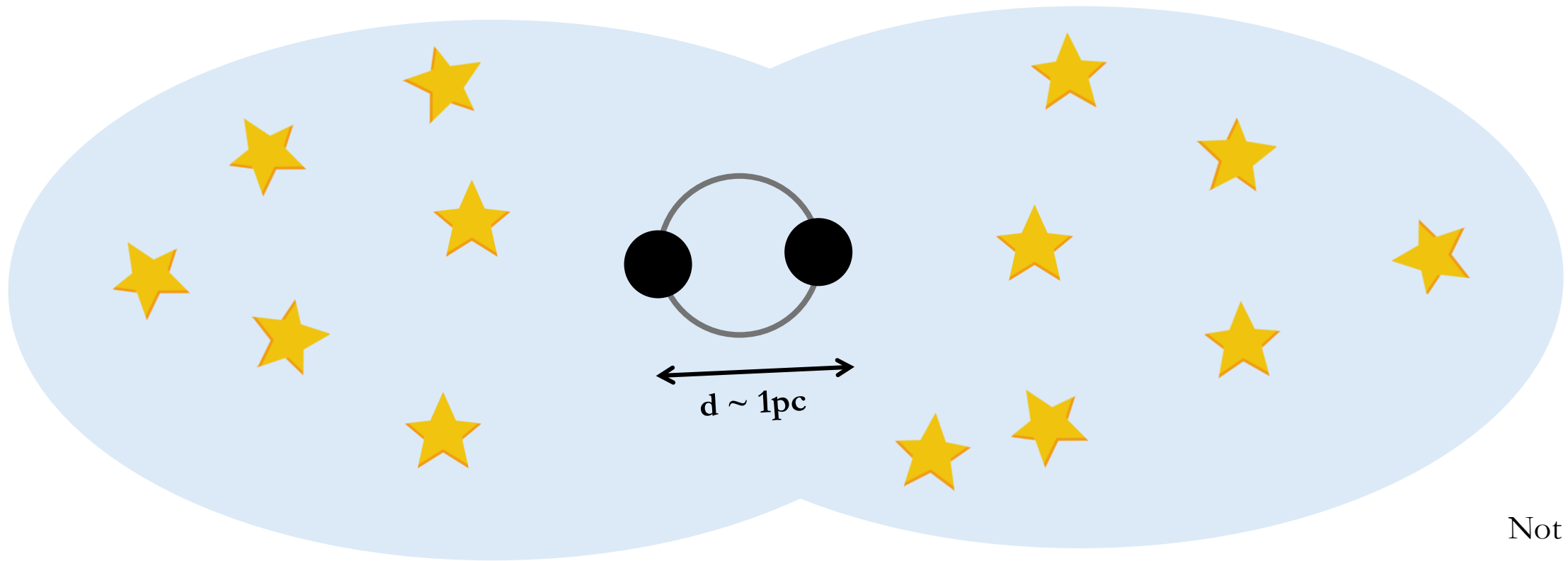


$$\frac{dL}{dt} = -\bar{M}_* \bar{v}_{ej} \frac{L^2}{GM^3} n_{int}$$

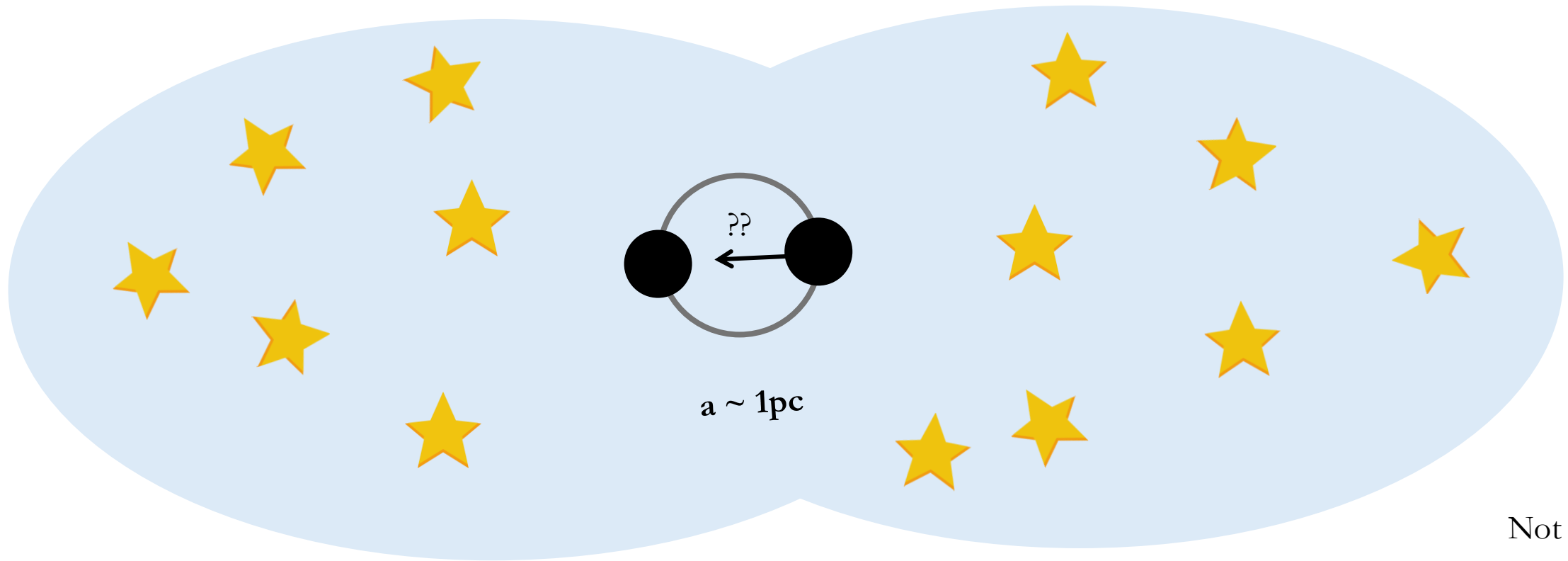
$$\Rightarrow t_0 = \frac{\sqrt{GM^3}}{\bar{M}_* \bar{v}_{ej} n_{int}} \left(\frac{1}{a_{min}} - \frac{1}{a_0} \right)$$

- At what binary separations (a_0) is this applicable?
- Recall the stellar density of the milky way
- Turns out: $n_{int} \propto \rho_* \sim 1 \text{ star/pc}^3$
- Once $a_0 \sim 1 \text{ pc}$, there are no more stars

Dynamical stellar interactions are good at getting the BHs to $d \sim 1\text{pc}$



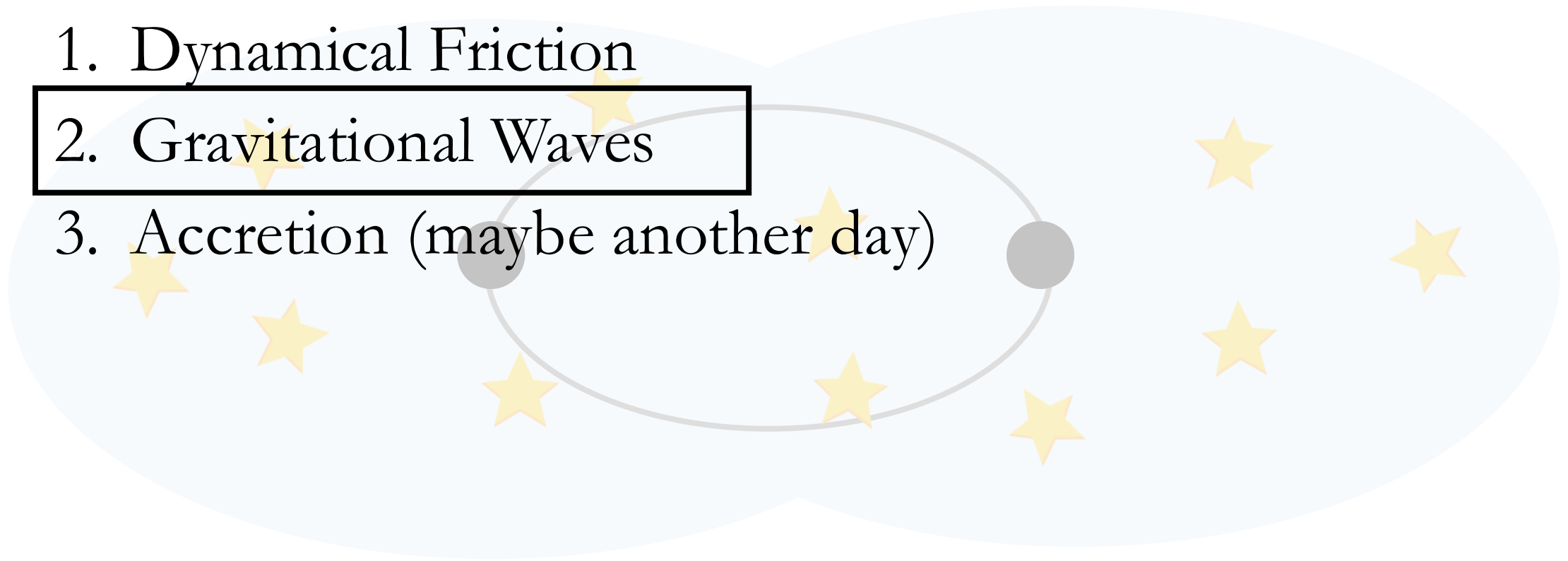
How do we overcome this final parsec?



Not to scale

Angular Momentum Extraction

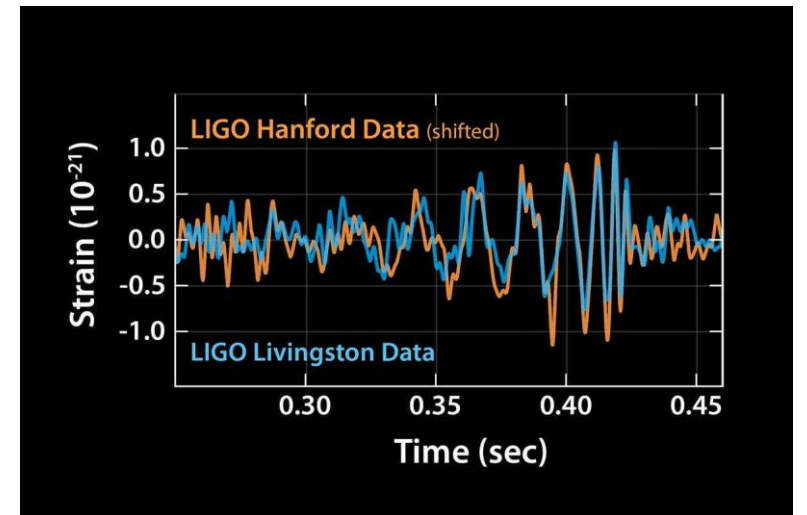
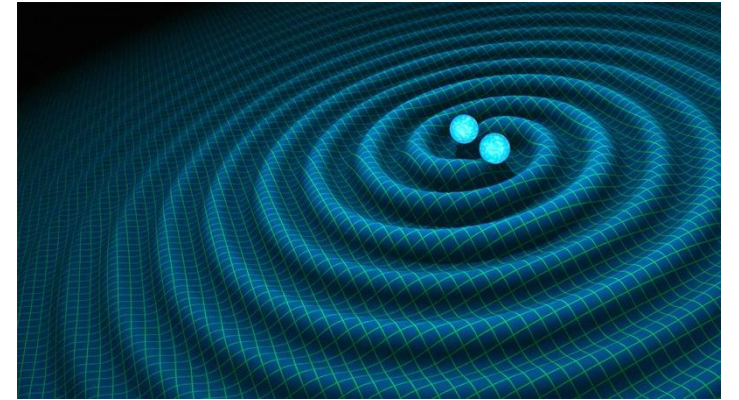
1. Dynamical Friction
2. Gravitational Waves
3. Accretion (maybe another day)



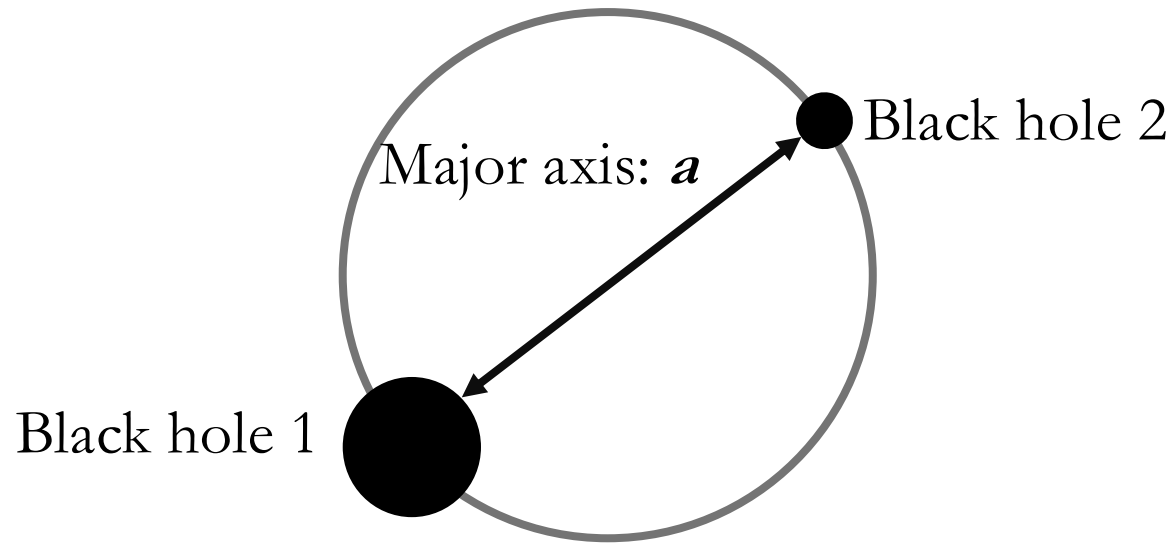
Gravitational Waves (GW)

Come up to the board, write or draw,
words or equations

- Gravitational waves are 'ripples' in space-time caused by some of the most violent and energetic processes in the Universe.
(LIGO)
- Direct consequence (and proof) of Einstein's theory of general relativity



Gravitational Waves (GW)

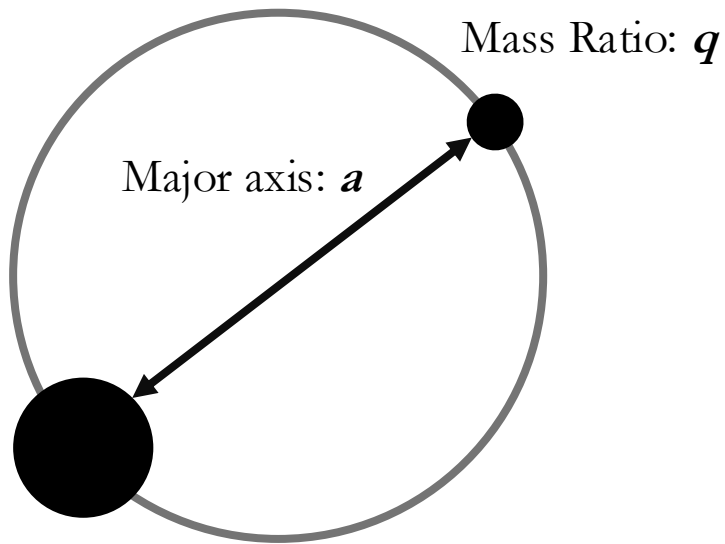


Binary Mass: M

Reduced mass: $\mu = M_1 M_2 / (M_1 + M_2)$

Mass Ratio: $q = M_2 / M_1$

GW Driven Inspiral



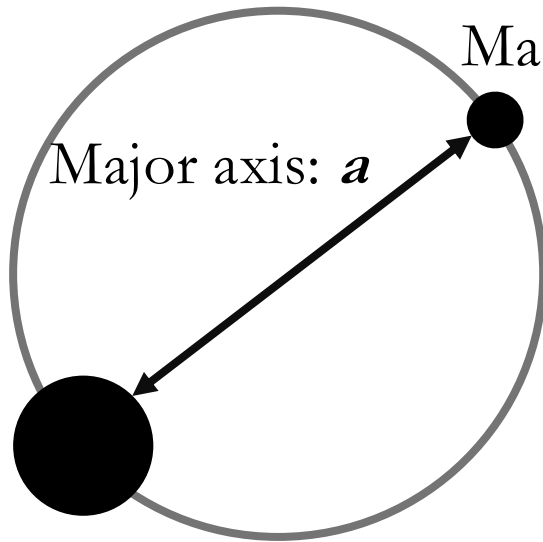
Binary Mass: M

Reduced mass: $\mu = M_1 M_2 / (M_1 + M_2)$

- Binary has a total orbital energy
- Gravitational waves emit energy, reducing the binary's total energy
- The binary begins to *inspiral*

GW Merger Timescale

- If the binary has major axis a
- We can define a da/dt due to GW emission
- Calculate a time scale for $a \rightarrow 0$



Binary Mass: M

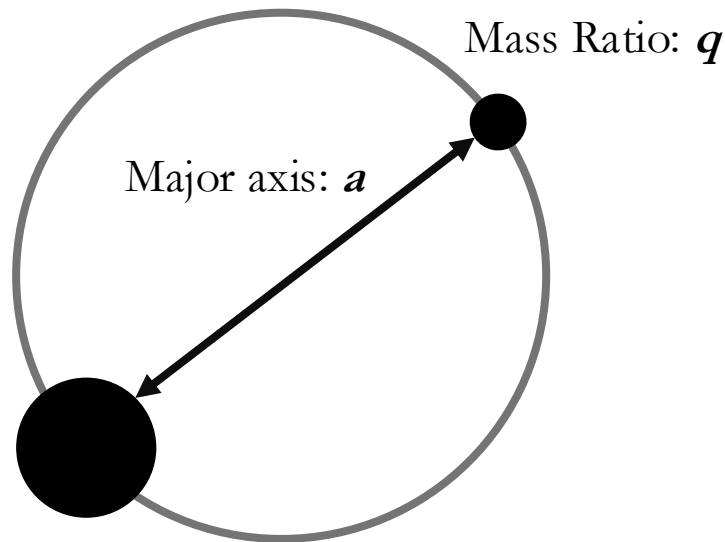
Reduced mass: $\mu = M_1 M_2 / (M_1 + M_2)$

Recall HW6 and Dynamical Friction:

$$\frac{d\vec{v}_M}{dt} = -\frac{4\pi \ln(\Lambda) G^2 \rho M}{v_M^2} \hat{v}_M$$

$$\Rightarrow T_{stop} = \frac{v_M^3}{12\pi \ln \Lambda G^2 \rho M}$$

GW Merger Timescale (your turn)



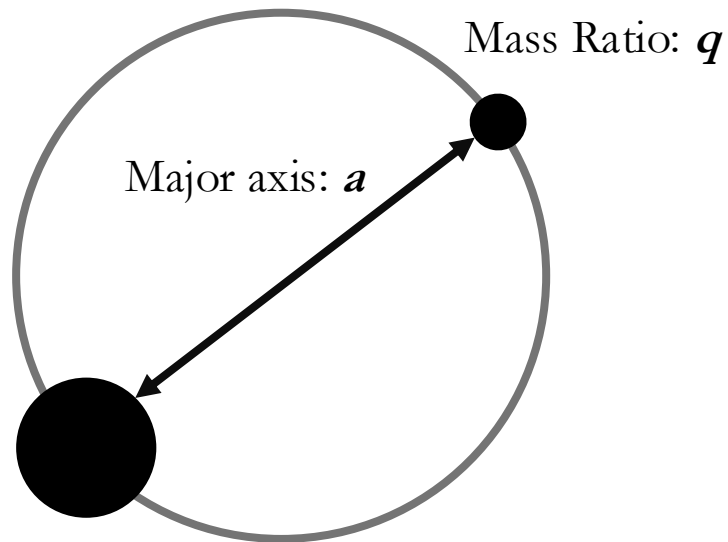
Binary Mass: M

Reduced mass: $\mu = M_1 M_2 / (M_1 + M_2)$

- If the binary has major axis a
- We can define a da/dt due to GW emission
- Your turn: Calculate a time scale for $a \rightarrow 0$
- When integrating, don't forget to define limits

$$\frac{da}{dt} = - \frac{1}{a^3} \frac{64 G^3 \mu M_1 M_2}{5 c^5} \quad = \text{some constant}$$
$$\Rightarrow t_0 = ??$$

GW Merger Timescale



Binary Mass: M

Reduced mass: $\mu = M_1 M_2 / (M_1 + M_2)$

$$\frac{da}{dt} = -\frac{1}{a^3} \frac{64 G^3 \mu M_1 M_2}{5 c^5}$$

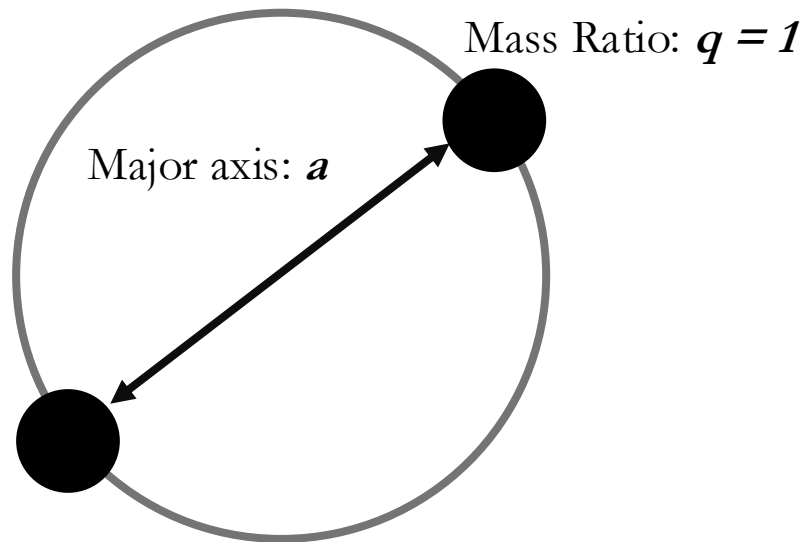
$$a^3 da = -\frac{64 G^3 \mu M_1 M_2}{5 c^5} dt$$

$$\int_{a_0}^0 a^3 da = -\int_0^{t_0} A dt$$

$$-\frac{1}{4} a_0^4 = -A t_0 \Rightarrow t_0 = -\frac{1}{4A} a_0^4$$

$$t_0 = \frac{5}{256} \frac{c^5}{\mu M_1 M_2 G^3} a_0^4$$

GW Merger Timescale



Binary Mass: $M = M_1 + M_2$

Reduced mass: $\mu = M_1 M_2 / (M_1 + M_2) = M/4$

$$t_0 = \frac{5}{256} \frac{c^5}{\mu M_1 M_2 G^3} a_0^4$$

$$t_0 = \frac{5}{16} \frac{c^5}{G^3} \frac{a_0^4}{M^3}$$

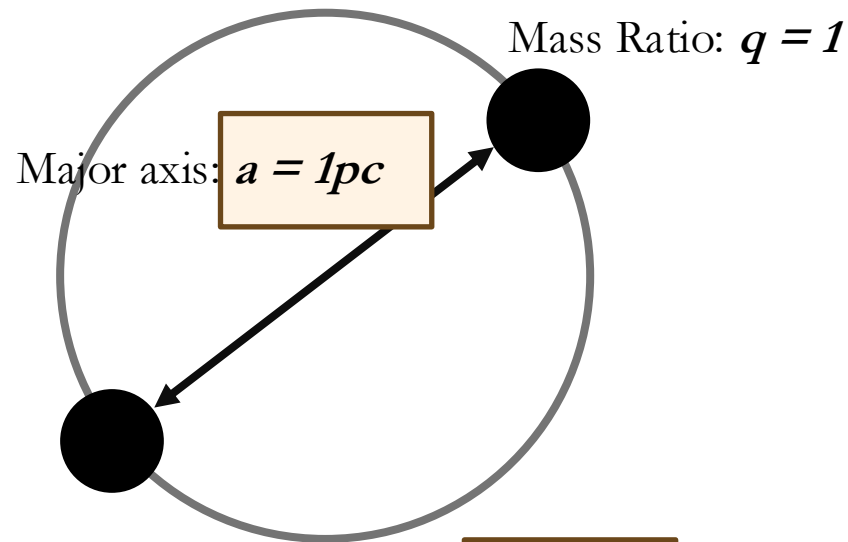
$$t_0 = \left(10^{31} \frac{M_\odot^3 \text{ Gyr}}{\text{pc}^4} \right) \frac{a_0^4}{M^3}$$

These are funky units, what happens if you let $a_0 \sim 1 \text{ pc}$ and $M \sim M_\odot$?

Calculate the merger time
for this binary

$$t_0 = \frac{5}{16} \frac{c^5}{G^3} \frac{a_0^4}{M^3}$$

$$t_0 \sim \left(10^{31} \frac{M_\odot^3 \text{ Gyr}}{\text{pc}^4} \right) \frac{a_0^4}{M^3}$$



Binary Mass: $M = M_1 + M_2 = 10^7 M_{\text{sun}}$

Reduced mass: μ

Plugging in the numbers to the left...

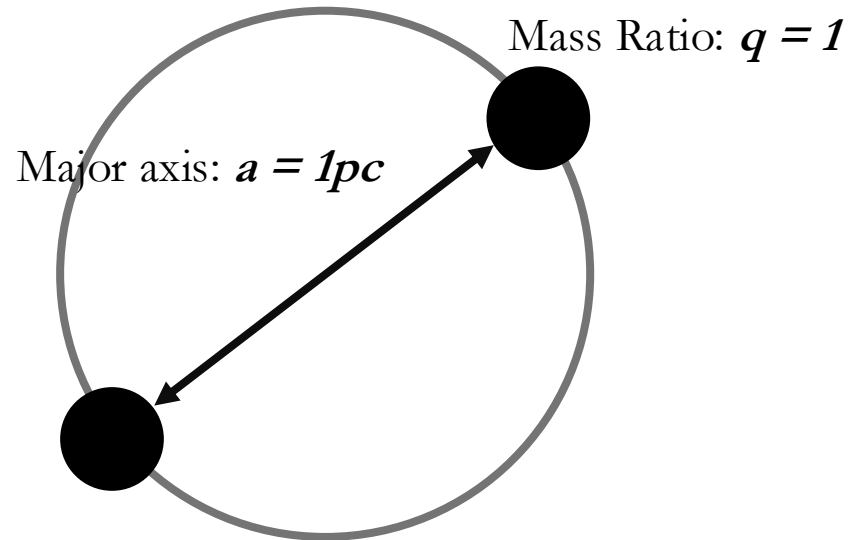
A. $t_0 \sim 10^{31} \text{ Gyr}$ B. $t_0 \sim 10^{24} \text{ Gyr}$

C. $t_0 \sim 10^1 \text{ Gyr}$ D. $t_0 \sim 10^{10} \text{ Gyr}$

$$t_0 \sim \left(10^{31} \frac{M_\odot^3 \text{ Gyr}}{\text{pc}^4} \right) \frac{(1pc)^4}{(10^7 M_\odot)^3}$$

The universe is only ~ 14 Gyr old!!

$$t_0 \sim 10^{10} \text{ Gyr}$$

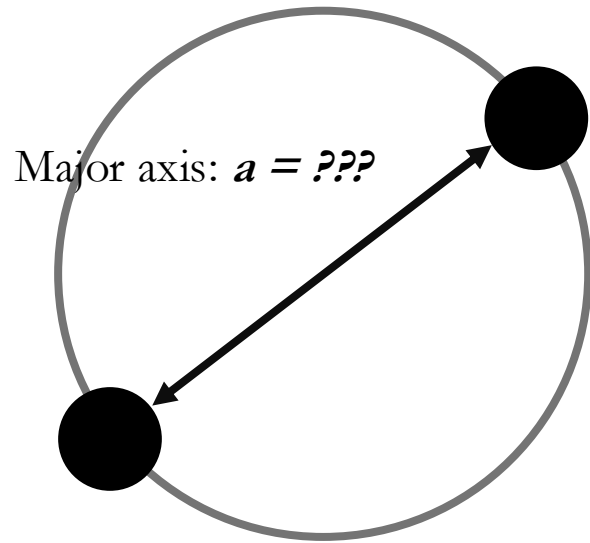


Binary Mass: $M = M_1 + M_2 = 10^7 M_{\text{sun}}$

Reduced mass: μ

Gravitational waves would take *too long* to drive the binary to merger from $a = 1pc$

At what separation are GWs efficient?



$$t_0 \sim \left(10^{31} \frac{M_{\odot}^3 \text{ Gyr}}{\text{pc}^4} \right) \frac{a_0^4}{M^3}$$

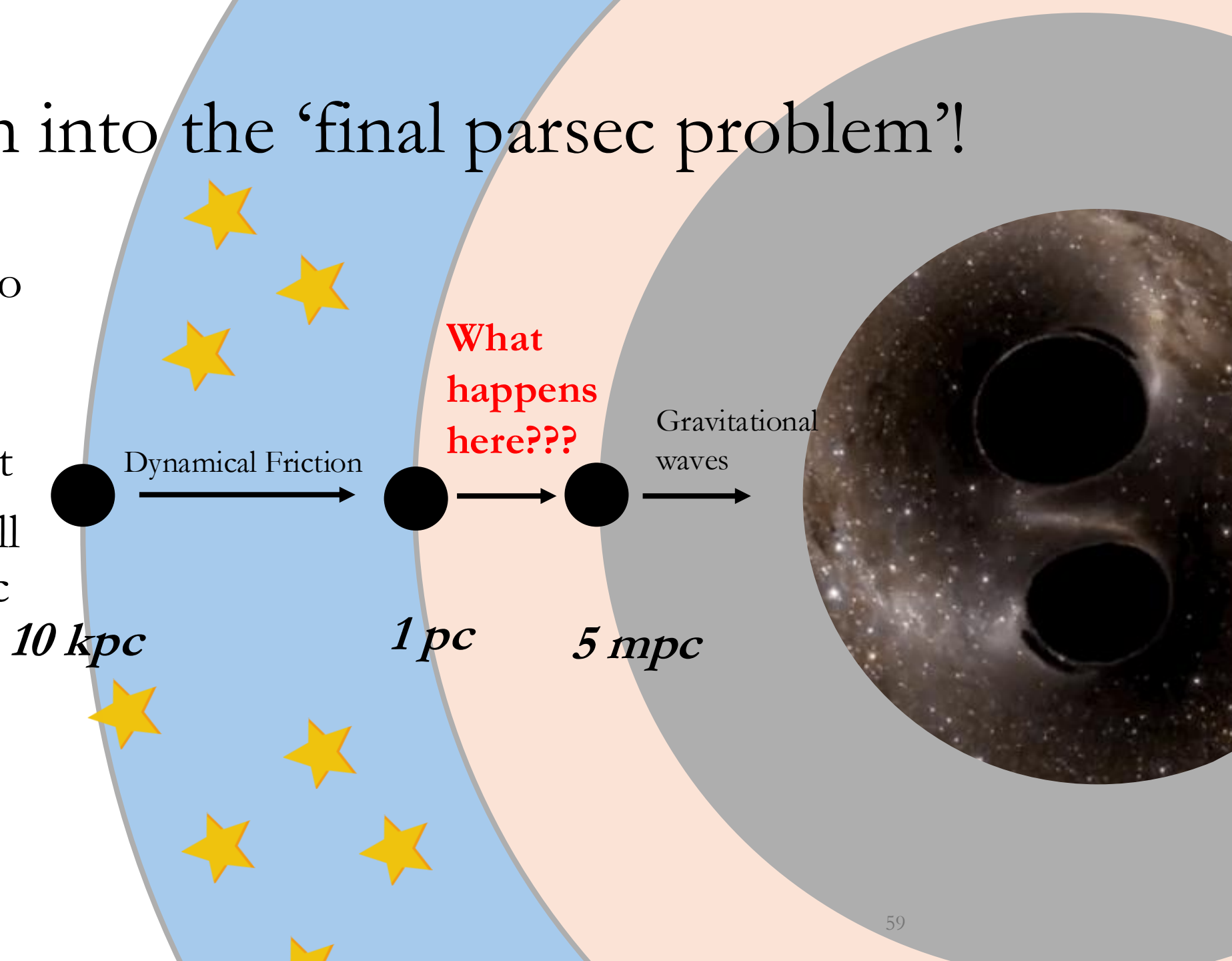
$$a_0^4 \sim \left(10^{-31} \frac{\text{pc}^4}{M_{\odot}^3 \text{ Gyr}} \right) t_0 M^3$$

Let $t_0 = 10 \text{ Gyr}$ (Hubble time):

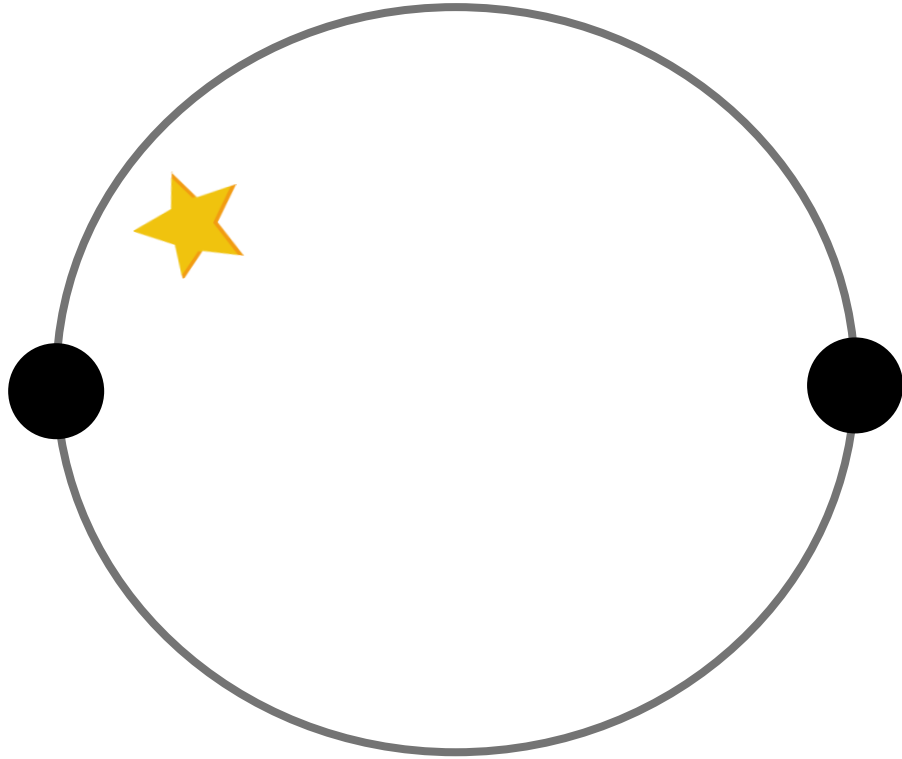
$$a_{\text{GW}} \sim 5 \text{ mpc} \left(\frac{M}{10^7 M_{\odot}} \right)^{3/4}$$

We have run into the ‘final parsec problem’!

- Not enough stars to extract angular momentum
- GWs are inefficient
- This is the very well known ‘final parsec problem’
(Milosavljevic & Merritt 2003)

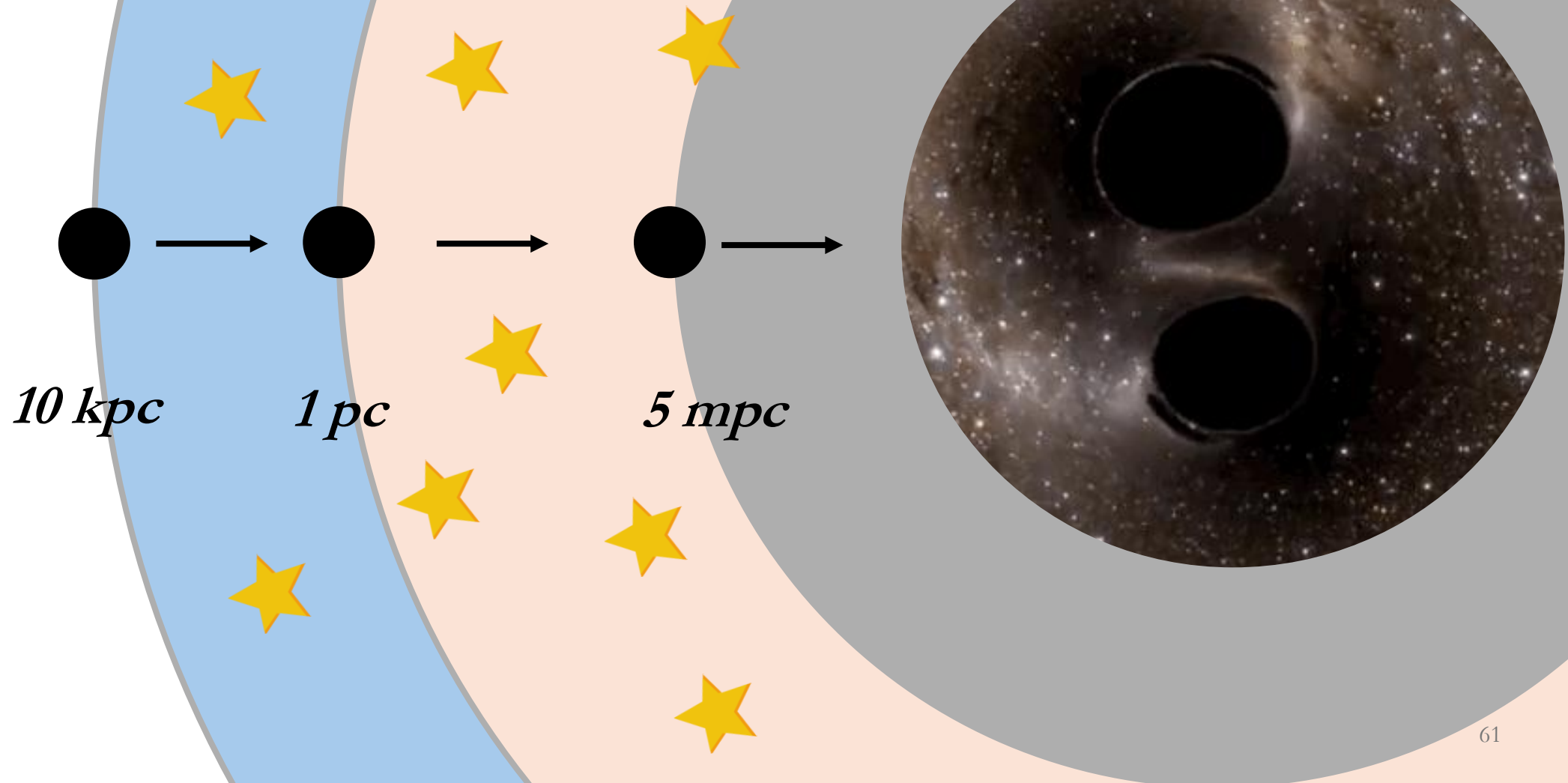


The ‘final parsec problem’ is solved!



- Weird galaxy shapes allow the loss cone to be refilled by different stellar orbits (Gualandris et al. 2016)
- There are enough stars for the SMBHs to eject and get closer together
- Different types of dark matter can even solve this problem (Alonso-Alvarez et al. 2024.)

Multiple stages and processes to SMBH inspirals



In class assignment

On Thursday

- Don't forget to submit your notes for the in-class assignment?
- Rest of today's lecture
- More about my research! What is the cutting-edge research today?
- Playing with gravitational waveforms – notebooks and open-source science